Frobenius product, parallel circuits, and sample-complexity barriers for deep Gaussian Bayesian networks

ChatGPT-6 Astra Kaarel Hänni Dmitry Vaintrob*

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Abstract

We prove that a Boolean function with a globally unit-margin neural representation of layer Frobenius product P has a nonuniform threshold circuit of size polynomial in $d + P$ and depth $O(\log(2 + P))$. The original network may have arbitrary finite depth and widths, and each neuron may use its own arbitrary zero-preserving 1-Lipschitz activation. Thus polynomial product implies nonuniform TC^1, and hence NC^2. The proof combines a two-ball covering lemma, finite-set sampling of a short prefix, and bounded-precision lookup tables; it does not require computable activations or well-conditioned intermediate layers.

For the original tanh Gaussian posterior conditioned only on a global absolute margin and exact training labels, we prove a sample-to-circuit converse uniformly at every finite scale $0 < \sigma^2 \leq 1/N$. The circuit resources are polynomial in input dimension, sample count, and learner depth. The theorem requires logarithmic depth padding and uses a width threshold exponential in input dimension. Its proof combines finite canonical features, Gaussian block covers, and factorially convergent parallel Picard iteration. The block covers keep the number of representatives polynomial even when the common prior prefactor approaches one. All confidence-failure tolerances $\varepsilon < 1/2$ are allowed. A separate above-critical product ceiling gives circuit resources independent of learner depth. Consequently, for any prescribed prior-scale schedule in the stated range, a universal sample bound polynomial jointly in teacher resources and learner depth would imply $\mathsf{P/poly} = \mathsf{TC}^1$. Conversely, the stronger assumption $\mathsf{P/poly} = \mathsf{TC}^0$ suffices for that joint polynomial learning bound, by compiling one universal circuit-evaluation family. An explicit finite-horizon compiler shows that the necessary collapse already follows for teachers of width 24 and weights bounded by one, while the learner width remains unrestricted. In the opposite direction, previously established learning bounds imply that any fixed stretched-exponential sample lower bound $\exp(\Omega(L^\alpha))$, $\alpha > 0$, with teacher resources polynomial in depth L would imply $\mathsf{NC} \neq \mathsf{P/poly}$. These are barriers to the two proposed resolutions, not a resolution of unrestricted-depth learning itself.

1 Introduction

A deep network can have small individual weights while its product of layer norms grows exponentially with depth. These two resource measures lead to different questions. A bound on width, depth, and individual weights gives a finite description of a globally margined Boolean classifier. A bound on the product of Frobenius norms imposes a stronger restriction: we show that it gives a small parallel circuit, regardless of the network's original depth.

The geometric reason is simple. A linear map of Frobenius norm M sends every Euclidean ball of radius r into the union of two balls of radius $Mr/\sqrt{2}$. Coordinatewise 1-Lipschitz maps preserve

*Authors are listed in alphabetical order.

this property. After $O(\log P)$ layers, a network of total product P has only polynomially many prefix states relevant to its output signs. The remaining layers can be replaced by a finite classification of these states. A separate sampling argument bounds the width of the retained prefix. Normalizing each layer before numerical simulation removes any dependence on the sizes of intermediate signals.

The result has consequences for Bayesian learning. In the original Gaussian tanh model, the prior is conditioned on $|z(x)| > 1/2$ at every Boolean input and on the observed labels. Above Gaussian exponent one, an existing finite-scale posterior ceiling places most posterior mass on Boolean functions with product polynomial in the input dimension and sample count, uniformly in learner depth. Combining that ceiling with our circuit theorem gives a converse: a distribution-free sample guarantee forces the target to have a small parallel circuit. Two finite majority constructions handle posterior mixtures and confidence-failure tolerances arbitrarily close to $1/2$.

The critical exponent one requires a different representation theorem. We construct finite canonical features preserving every query's correlations with a common anchor set, without preserving the entire kernel. Their dimension grows additively with depth. The resulting recursion is a linear contraction sequence plus residual maps of small total Lipschitz constant, and parallel Picard iteration evaluates it in logarithmic threshold depth. Uniform importance quadrature makes each map a finite circuit despite arbitrary controlled densities and ill-conditioned anchor spans. A finite-event entropy certificate then gives an actual posterior ceiling on the full cube. The resulting converse includes L in its polynomial resources, assumes $L \geq C \log(d + n + L + 2)$, and chooses width above $2^{Cd}\,\mathrm{poly}(d + n, L)$. These qualifications still permit the universal polynomial-bound implication below.

Fixed prefactors $0 < t < 1$ need one more argument, since their exponent gaps shrink with width. For the two largest eigenvalues of a controlled row's second moment, Gaussian entropy bounds its two-ball covering radius more strongly than its ordinary Lipschitz gain. The strict gap accumulates over a short prefix while charging the entire suffix. A normalized importance rule evaluates that prefix without representing large or small physical scales. This first gives a converse uniform on each range $0 < t \leq t_0 < 1$. To cover the moving boundary $t \to 1$, we group $O((t^{-2} - 1)^{-1})$ layers at a time. On most blocks the linear map is a contracting isometry plus a Hilbert–Schmidt perturbation of bounded size, and the nonlinear residual is small. Each such block admits a constant-size cover. Only logarithmically many blocks are needed, while parallel Picard iteration evaluates their potentially long prefix. The final theorem is uniform over every $0 < t \leq 1$, equivalently every $0 < \sigma^2 \leq 1/N$.

There are two separate consequences. A universal joint polynomial sample upper bound would collapse P/poly to nonuniform TC^1. The stronger collapse $\mathsf{P}/\mathrm{poly} = \mathsf{TC}^0$ is sufficient for that learning bound: one universal evaluator fixes the resource exponent for all bounded-weight teachers. Any fixed stretched-exponential depth lower bound, when both input dimension and bounded-weight teacher resources are polynomial in depth, would separate P/poly from every nonuniform polylogarithmic-depth Boolean circuit class. Neither consequence is currently an unconditional complexity theorem. The arguments also leave room for intermediate sample growth between polynomial and exponential. Appendix A shows that the necessary collapse already applies to fixed-width teachers: a width-24 tanh network with coefficients bounded by one can simulate any small Boolean circuit at polynomially larger depth. Its memory is exponentially delicate, so this observation does not supply a robust or polynomial-product representation.

The geometric, canonical-feature, finite-event extraction, and circuit reductions are proved below. We state the previously established inputs explicitly: the finite-scale posterior ceiling in Section 5, the scalar Bellman facts and finite-width Fourier target mass in Section 7, the companion scale-change target mass in Section 8, and the circuit-target learning bound in Section 11. The critical endpoint is proved separately, rather than inferred by taking a limit in the above-one result.

The conditional sufficiency direction also uses the companion normalized intrinsic learning theorem, stated explicitly where needed.

Relation to previous work. Norm-controlled generalization and compression are well established. Golowich, Rakhlin, and Shamir [7] bound Rademacher complexity under layer norm constraints, improving depth dependence and obtaining size-independent bounds under additional assumptions. Their full-version Theorem 3 already obtains a rank-one layer replacement and a Lipschitz suffix; its displayed uniform error decreases polynomially with the retained prefix length. Here the global Boolean margin is preserved using a logarithmic prefix and polynomially many covering states. Sellke [8] refines the Frobenius-product analysis for ReLU networks, often removing explicit depth dependence. Our sampling and normalization arguments belong to this broader norm-compression framework. The conclusion studied here is an exact nonuniform circuit for a globally margined Boolean classifier, followed by a comparison with the original conditioned Bayesian posterior. A norm-based generalization estimate alone does not make that posterior-to-teacher-complexity comparison.

Circuit-complexity barriers to neural depth separation also precede this work. Vardi and Shamir [9] relate deeper ReLU approximation lower bounds to circuit lower bounds and natural-proof barriers, using a reduction to polynomially bounded weights. Vardi, Reichman, Pitassi, and Shamir [10] establish related barriers for approximating efficiently computable functions with polynomial Lipschitz bounds. The implications here concern a different quantitative object: distribution-universal posterior sample complexity for bounded-resource tanh Boolean teachers. We make no claim that connecting neural representation bounds to circuit complexity, or compressing norm-bounded networks, is new.

Jacot [11, 12] studies a related depth baseline for homogeneous networks: the leading representation cost is governed by a nonlinear rank, with finite-depth corrections and quantitative bounds on deviation from low-rank layer matrices. Those results help distinguish low-dimensional representation from the complexity of its encoder and decoder. They do not by themselves bound that residual complexity by a polynomial in the teacher resources considered here.

2 Networks and circuit conventions

Let $X_d = \{-1,1\}^d$ and $f : X_d \to \{-1,1\}$. A network of weighted depth $D \geq 1$ has augmented input $h^0(x) = (1, x)$, hidden layers

$$h_i^s(x) = \phi_{s,i}\big((W^s h^{s-1}(x))_i\big), \qquad 1 \leq s < D,$$

and scalar linear output $z(x) = W^D h^{D-1}(x)$. Every hidden width is finite. Unless specified otherwise, each $\phi_{s,i} : \mathbb{R} \to \mathbb{R}$ satisfies

$$\phi_{s,i}(0) = 0, \qquad |\phi_{s,i}(u) - \phi_{s,i}(v)| \leq |u - v|.$$

There are no additional affine biases: constants enter through the augmented input and may be propagated by ordinary hidden units. Write

$$P_F(W) = \prod_{s=1}^{D} \|W^s\|_F .$$

A global unit-margin representation means $f(x)z(x) \geq 1$ for every $x \in X_d$. We allow arbitrary real weights and arbitrary activations in this definition; no computability assumption is made.

All circuit classes in this paper are *nonuniform*. Boolean circuits use a fixed bounded-fan-in basis, for example $\{\mathsf{AND}, \mathsf{OR}, \mathsf{NOT}\}$, together with constants. Threshold circuits may equivalently use unbounded-fan-in majority gates and bounded-fan-in Boolean gates. Our constructed weighted threshold gates have polynomially bounded integer weights and can be converted to majority gates with polynomial overhead. Thus TC^1 denotes polynomial size and logarithmic depth in the ordinary nonuniform sense. We use

$$\mathsf{NC} = \bigcup_{k \geq 1} \mathsf{NC}^k,$$

where NC^k has polynomial size and depth $O(\log^k d)$. Size may count wires and input nodes; retaining all d named input nodes changes our bounds only by an additive d.

Theorem 2.1 (Product-to-parallel-circuit theorem)**.** *There are universal constants $a, c > 0$ such that the following holds. If $f : X_d \to \{-1, 1\}$ has a global unit-margin representation as above with $P_F(W) \leq P$, then $P \geq 1$ and f has a threshold circuit with at most $(d + P + 2)^a$ wires and depth at most $c \log(P + 2)$. It also has a bounded-fan-in Boolean circuit of polynomial size in $d + P$ and depth at most $c \log^2(P + 2)$, after changing a, c.*

The constants are independent of the original depth, widths, physical weights, and choices of activations. In particular, any family with $P = \mathrm{poly}(d)$ belongs to nonuniform $\mathsf{TC}^1 \subseteq \mathsf{NC}^2$.

The theorem is about represented functions. It does not give an efficient procedure for finding a circuit from a network, and does not put the evaluation problem for supplied arbitrary real networks in NC. Its proof deliberately permits auxiliary activations outside the fixed-tanh class.

3 A short prefix suffices

3.1 Relevant inputs

Lemma 3.1 (Relevant-coordinate reduction)**.** *Every globally unit-margin representation has Frobenius product at least one. Its Boolean function depends on at most $P_F(W)^2$ input coordinates. After fixing the other coordinates, it has a representation on just the relevant coordinates whose product is no larger.*

Proof. Zero preservation and the Lipschitz bounds imply

$$\mathbb{E}_{x \sim \mathrm{Unif}(X_d)} z(x)^2 \leq \left(\prod_{s=2}^{D} \|W^s\|_F^2 \right) \mathbb{E} \left\| W^1 (1, x) \right\|_2^2 = P_F(W)^2.$$

Here $\mathbb{E}[(1, x)(1, x)^T] = I$, and the formula also covers $D = 1$. The margin makes the left side at least one.

Put $T = \prod_{s=2}^{D} \|W^s\|_F$, with an empty product equal to one. For each relevant coordinate i, some pair of inputs differing only at i has opposite labels. Their outputs differ by at least two, so

$$2 \leq 2T \left\| W^1 e_i \right\|_2 .$$

Summing the resulting squared inequalities over relevant coordinates gives their number $k \leq P_F(W)^2$.

For irrelevant coordinates choose independent uniform signs a_i. Absorb them into the first bias column by replacing w_0 with $w_0 + \sum_i a_i w_i$, and delete their columns. The expected squared Frobenius norm of the new first matrix equals the old squared norm. Some choice therefore does not increase it. Irrelevance ensures that all these restrictions preserve the Boolean function and its margin. □

3.2 The two-ball lemma

Lemma 3.2 (Dimension-free two-ball covering)**.** *For every finite real matrix W, center a, and $r \geq 0$, and every coordinatewise 1-Lipschitz map Φ, the set $\Phi(WB(a,r))$ is covered by at most two Euclidean balls of radius $r\|W\|_F/\sqrt{2}$.*

Proof. Zero radius and the zero matrix are immediate. Translation and scaling reduce the linear part to $WB(0,1)$ with $\|W\|_F = 1$. Let α be its largest singular value. If $\alpha^2 \leq 1/2$, the operator-norm bound gives one ball. Otherwise put

$$b^2 = 1 - \alpha^2, \qquad c = \frac{\alpha^2 - b^2}{2\alpha} = \frac{2\alpha^2 - 1}{2\alpha},$$

and let u be a leading left singular vector. The remaining singular part has operator norm at most b. For a unit-ball input whose leading right-singular coordinate is t, distance squared from its image to the center cu or $-cu$ matching the sign of t is at most

$$(\alpha|t| - c)^2 + b^2(1 - t^2).$$

This is convex as a function of $|t| \in [0,1]$. Both endpoint values are

$$c^2 + b^2 = (\alpha - c)^2 = \frac{1}{4\alpha^2} \leq \frac{1}{2}.$$

The two centers therefore give the claimed linear cover. Apply Φ to the translated and rescaled centers; its Euclidean Lipschitz constant is at most one. The centers need not belong to the image, which does not affect the covering assertion. □

Lemma 3.3 (Margin-relevant prefix states)**.** *Let a network of product P have global unit margin on a set of inputs of Euclidean norm at most R. Put $S_j = \prod_{s \leq j} \|W^s\|_F$. For $j \leq D-1$, its jth hidden image has a cover by at most 2^j balls of radius*

$$r_j = RS_j 2^{-j/2}.$$

If

$$j = \lceil 2\log_2(16PR) \rceil \leq D - 1,$$

one can select at most $512P^2R^2$ actual normalized prefix states $c_a = h^j(x_a)/S_j$, each with its label $f(x_a)$, so that every $u = h^j(x)/S_j$ has a same-label selected state within $1/(8P)$, while every selected state of the opposite label is at distance at least $2/P$. All these states have norm at most R.

Proof. Iterate Lemma 3.2, starting from the radius-R input ball. The remaining suffix is Lipschitz with constant at most P/S_j. Two hidden images in one covering ball therefore have output difference at most

$$2r_jP/S_j = 2PR2^{-j/2} \leq 1/8,$$

whereas opposite labels require difference at least two. Each nonempty ball is thus monochromatic. Select an actual image in each nonempty ball; this doubles the cover radius. Divide by S_j to get the same-label bound. The suffix Lipschitz bound gives the opposite-label bound directly. Finally $2^j \leq 2(16PR)^2$, and zero preservation gives $\|h^j(x)\| \leq S_jR$. □

3.3 Sampling only the retained prefix

We next bound the width of the short prefix. The following argument uses the freedom to change the neuronwise activations; it is not a fixed-activation tanh compression claim.

Lemma 3.4 (Finite-set prefix compression)**.** *Let X be a nonempty finite set of inputs of norm at most R, and let a network as in Section 2 have product at most P. For $1 \le h \le D-1$ and $\epsilon > 0$, its first h hidden layers can be replaced by layers of width at most*

$$m = \left\lceil \max\{1, 64(PRh/\epsilon)^2(1 + \log|X|)\} \right\rceil$$

so that all final outputs on X change by at most ϵ and the full product never increases. The replacement activations remain zero-preserving and 1-Lipschitz. No bound depends on the number or widths of the later layers.

Proof. Consider consecutive matrices W, V. Let w_i be row i of W and v_i column i of V, and put $a_i = \|w_i\|_2$ and $b_i = \|v_i\|_2$. Terms with $a_i b_i = 0$ contribute zero and may be deleted. If all contributions are zero, the intervening computation can be replaced by a zero layer. Otherwise set

$$A = \sum_i a_i b_i \le \|W\|_F \|V\|_F =: B, \qquad p_i = \frac{a_i b_i}{A}, \qquad u_i = \frac{w_i}{a_i}, \qquad \psi_i(t) = \frac{\phi_i(a_i t)}{a_i}.$$

The activation ψ_i has the required properties. The preactivation of V is $\sum_i a_i v_i \psi_i(u_i^T h)$. Sample m indices independently with probabilities p_i. For each sampled index use incoming row u_i, activation ψ_i, and outgoing column $Av_i/(mb_i)$. The new preactivation is an unbiased empirical average of the random vectors

$$Y(h) = \frac{Av_i}{b_i}\psi_i(u_i^T h).$$

For every draw, the two new Frobenius norms are $\sqrt{m}$ and $A/\sqrt{m}$, so their product is $A \le B$.

Let S be the product of all preceding layer norms. On X one has $\|h\| \le SR$, hence $\|Y(h)\| \le ASR =: J$. The finite-input Hilbert sampling estimate gives a common sample with every preactivation error smaller than δ whenever

$$64J^2(1 + \log|X|) \le m\delta^2, \qquad m \ge 1.$$

Here is a finite-sum proof with deliberately loose constants. If F_x is the norm of the centered sum of m sampled vectors, the Hilbert variance identity gives $\mathbb{E}F_x \le J\sqrt{m}$. Changing one index changes F_x by at most $2J$. A finite bounded-differences argument, obtained by iterating the scalar bound $\mathbb{E}e^{t(Z-\mathbb{E}Z)} \le e^{t^2c^2/2}$ for a variable of oscillation at most c, gives

$$\mathbb{E}e^{t(F_x - \mathbb{E}F_x)} \le e^{2t^2 m J^2}.$$

The scalar bound follows from convexity of the exponential and $\cosh u \le e^{u^2/2}$. Summing these exponential potentials over X and taking $t = \delta/(8J^2)$ proves the displayed sampling estimate: its budget bounds the mean by $m\delta/2$ and makes the total exponential potential strictly smaller than $e^{tm\delta/2}$. Thus some common sample has $F_x < m\delta$ for every x. This argument includes $|X| = 1$. If $J = 0$, all sampled features vanish. A zero full product makes the final network output identically zero. Both cases are handled directly.

When the current product and SB are positive, the suffix after V, including its activation if present, is Lipschitz with constant at most $P/(SB)$. Take $\delta = (\epsilon/h)SB/P$. Since $A \le B$, the displayed choice of m guarantees that the final output error in this replacement is at most ϵ/h.

Zero-product networks are replaced by zero networks before making any division. Process the first h hidden layers successively, allotting error ϵ/h to each. The product does not increase at any step, and altering the next matrix never changes an already bounded hidden width. Summing the output errors proves the assertion. □

4 Compiling the prefix at bounded precision

Proof of Theorem 2.1. Apply Lemma 3.1. If f is constant, a constant gate suffices. Otherwise retain $k \le P^2$ relevant inputs and put $R = \sqrt{k+1}$. Let $\overline{P} = 2P$ and

$$j_0 = \left\lceil 2\log_2(16\overline{P}R) \right\rceil, \qquad h = \min\{j_0, D-1\}.$$

Apply Lemma 3.4 with error $1/4$ to these h layers (nothing is required if $h = 0$), then double the final readout. The new network has signed margin at least $3/2$ and product at most $\overline{P}$. Its retained widths are at most a universal polynomial $C = C(P)$, since $R \le \sqrt{P^2+1}$, $1 + \log|X_k| = 1 + k\log 2$, and $h = O(\log(P+2))$.

If $D - 1 \ge j_0$, apply Lemma 3.3 to the new network using the upper bound $\overline{P}$. It supplies polynomially many normalized centers with the stated separation. If $D - 1 < j_0$, we will evaluate the entire short network and its linear readout instead. In both cases only $h \le j_0$ hidden layers need be simulated.

Write $M_s = \|W^s\|_F$, $S_s = \prod_{i \le s} M_i$, and $A_s = W^s/M_s$ for the new network. All M_s are nonzero because its margin is nonzero. Define $u_s = h^s/S_s$ and, neuron by neuron,

$$\Psi_{s,i}(v) = \frac{\phi_{s,i}(S_s v)}{S_s}.$$

Then the exact normalized computation is

$$u_s = \Psi_s(A_s u_{s-1}), \qquad \|A_s\|_F = 1, \qquad \|u_s\|_2 \le R.$$

Every $\Psi_{s,i}$ is zero-preserving and 1-Lipschitz, even when S_s is arbitrarily large or small.

Put $m_0 = \max\{C, k+1, 2\}$. Choose positive dyadic η, δ, each within a factor two of and no larger than the corresponding value

$$\frac{1}{4096\overline{P}(R+1)(j_0+1)}, \qquad \frac{1}{4096\overline{P}m_0(j_0+1)}.$$

Round each A_s entry on a dyadic mesh between $\eta/(2m_0)$ and η/m_0, so its Frobenius error is at most η. Set $B_0 = \lceil R + 2 \rceil$. Replace each $\Psi_{s,i}$ on $[-B_0, B_0]$ by a lookup table with grid spacing δ and table values rounded to multiples of δ. Lipschitzness gives scalar error at most 2δ. Its table length, all inverse meshes, and all integer ranges are polynomial in $P + 2$; every stored number has $O(\log(P+2))$ bits. This observation uses only existence of the finite table, not computability of the activation.

If e_s is the Euclidean error of the simulated state, then

$$e_s \le (1+\eta)e_{s-1} + \eta R + 2\sqrt{C}\,\delta, \qquad e_0 = 0.$$

The displayed choices give $e_s < 1/(64\overline{P})$ for every $s \le h$: for example, $(1+\eta)^{j_0} < 2$ and the accumulated right-hand error is at most $6/[4096\overline{P}]$. Rounded preactivations stay inside $[-B_0, B_0]$, as required.

In the long-network case, round each normalized center to Euclidean error at most $1/(64\overline{P})$, using a comparable dyadic mesh in each coordinate. A correct-label center then has computed distance at most $5/(32\overline{P})$, whereas every opposite-label center has computed distance at least $63/(32\overline{P})$. Choose a dyadic number

$$\tau \in [1/(2\overline{P}^2), 1/\overline{P}^2].$$

Compute the squared distances of the rounded vectors exactly, and output $+1$ precisely when some positive-labeled center has squared distance at most τ. A positive input has such a center, and a negative input has none. The gap proves exact classification. In the short-network case, the normalized final readout has norm one and signed margin at least $1/\overline{P}$. Round its coefficients to the same Frobenius precision and take its sign. Its additional error is at most $\eta R + (1+\eta)e_h < 1/(2\overline{P})$, so its sign is also exact.

It remains to justify the circuit bounds without an unproved transcendental-arithmetic assumption. All table values and rounded states have a common dyadic grid of inverse-polynomial spacing. All rounded matrix entries also have polynomially bounded dyadic denominators. After clearing denominators within one layer, an affine preactivation is a linear function of the preceding state bits with polynomially bounded integer weights and polynomial integer range. Compare this linear function to every table grid boundary by one threshold layer. Boolean gates identify its grid interval, and, for each output bit, an OR over those intervals whose stored table value has that bit equal to one implements the lookup. This has constant threshold depth and polynomial size. The lookup resets the output to its fixed grid, so denominators do not accumulate through the layers. Signed values can be encoded by an offset integer; this only changes the affine constants.

For distance tests, a difference of two rounded coordinates still has $O(\log(P+2))$ bits. Its square is a polynomial-size table. A threshold comparison of the sum of these squares implements each test, and an OR combines the positive tests. The short-case readout is itself a threshold gate. Every threshold used here has integer weights and a threshold of polynomial magnitude. Negative weights can be moved to complemented inputs and the constant threshold adjusted. Repeating inputs according to the positive integer weights, and adding constant inputs, converts such a gate to a majority gate with polynomial overhead. There are $O(\log(P+2))$ simulated layers, proving the threshold-circuit assertion.

For completeness, majority on q bits has a bounded-fan-in Boolean circuit of polynomial size and depth $O(\log(q+2))$. Use constant-depth full adders to compress three summands to two, in parallel at each bit position. After $O(\log(q+2))$ compression stages, two binary integers remain; add and compare them in $O(\log(q+2))$ further depth, even by a ripple circuit on their $O(\log(q+2))$ bits. Replacing each majority gate of our threshold circuit this way gives total depth $O(\log^2(P+2))$ and polynomial size. Finally, reconnect the retained input coordinates to their original names. This proves the theorem. □

Remark 4.1 (What the representation theorem does not assert). The proof bounds circuit resources, not the original network's parameters. It does not assert that a small bounded-weight deep network has polynomial product, or that a majority of low-product functions has low product. No closure of product norm under majority will be needed below. The arbitrary-activation compression is used only as an auxiliary step in compiling the represented Boolean function; the statistical model remains exactly tanh.

Corollary 4.2 (Other Lipschitz constants). *If the activations in hidden layer s are zero-preserving and have Lipschitz constants at most $\lambda_s > 0$, the conclusion of Theorem 2.1 holds with the effective*

product

$$P_{\text{eff}} = P_F(W) \prod_{s=1}^{D-1} \lambda_s$$

in place of P. In particular, the result includes ReLU.

Proof. Write $\phi_{s,i} = \lambda_s \psi_{s,i}$ with $\psi_{s,i}$ 1-Lipschitz. Replace the hidden state by h^s/λ_s, multiply the following matrix by λ_s, and use $\psi_{s,i}$ as the activation. The resulting representation has the same output and product P_{eff}. Apply Theorem 2.1. □

5 The original Gaussian posterior and its ceiling

Fix input dimension $d \geq 1$, weighted depth $L \geq 3$, and common hidden width N. In this section all hidden activations are tanh. The raw Gaussian prior has independent entries with variances

$$W_{ij}^1 : \frac{t^2}{d}, \qquad W_{ij}^s : \frac{t^2}{N} \ (2 \leq s < L), \qquad v_i : \frac{t^2}{N^2},$$

where $v = W^L$ is the linear readout. Equivalently, put $\sigma^2 = t^2/N$ and use energy coordinates with squared norm

$$\kappa_N(W) = \frac{d}{N} \left\| W^1 \right\|_F^2 + \sum_{s=2}^{L-1} \|W^s\|_F^2 + N \|v\|_2^2 .$$

Their coordinates have common variance σ^2. The exponent convention $\sigma^2 = N^{-\gamma}$ corresponds to $t^2 = N^{1-\gamma}$.

For a consistently labeled sample $S = ((x_i, y_i))_{i=1}^m$, let Q_S be the raw prior conditioned on

$$|z(x)| > 1/2 \quad (x \in X_d), \qquad y_i z(x_i) > 0 \quad (1 \leq i \leq m).$$

There are no further constraints on hidden activations, internal margins, operator norms, or parameter size. Let

$$p_S(f(x) \mid x) = Q_S\{f(x)z(x) > 0\}.$$

The confidence-failure mass under an input law $\mathcal{D}$ is $\mathcal{D}\{x : p_S(f(x) \mid x) \leq 2/3\}$.

We quote the following established result with its explicit constants. It is the finite-scale posterior energy ceiling in [1, section "A posterior energy ceiling and a necessary representation condition"]; the exact source file is identified in the bibliography.

Theorem 5.1 (Imported finite-scale posterior ceiling)**.** *Fix $d, n \geq 1$, $L \geq 3$, $\zeta > 0$, and $0 < \beta < 1$.*

Define

$$
\begin{aligned}
C_0 &= n+1,\\
R_0 &= \max\{4(n+1),(d+1)\log(16(n+1))\},\\
M &= \left\lceil \max\{4, C_0, 2R_0, dC_0(d+1)R_0^2\} \right\rceil, \qquad T = MC_0R_0^2,\\
K &= 2+(L+14)(2T)^{1/(L-1)},\\
b^2 &= \max\{2(d+1)\log 2 + 4\log 8,\ 16\log 9 + 8\log(8L)\},\\
A &= \left(\frac{K+2}{L-1}\right)^{(L-1)/2},\\
\xi &= \min\left\{1, \frac{1}{4\sqrt{L(K+1)}}, \frac{1}{8\sqrt{2}LA}\right\},\\
N_0 &= \left\lceil \max\left\{M, 2, (b/\xi)^{2/\zeta}, 2\log(2/\beta)\right\}\right\rceil.
\end{aligned}
$$

For every integer $N \geq N_0$, every actual finite scale $0 < \sigma^2 \leq N^{-(1+\zeta)}$, and every consistently labeled sample of at most n inputs, Q_S assigns probability at least $1-\beta$ to Boolean functions that admit a global unit-margin tanh representation of product at most P_0, where

$$\boxed{P_0^2 = d^{-1}e^{19}(2T)^{3/2}.}$$

Such a representation can be taken at the same width and depth. The width threshold N_0 is polynomial in $d, n, L, \log(2/\beta)$, with degree depending only on ζ.

The theorem concerns the Boolean function of a posterior draw. Its low-product representative need not be the draw itself. In particular, it would be incorrect to replace this result by an assertion that all posterior parameters have small energy.

Lemma 5.2 (Polynomial size of the ceiling)**.** *For $S = d+n+2$, the preceding product satisfies*

$$P_0 \leq e^{19/2}544^{3/4}S^9.$$

In particular its polynomial degree is independent of L, ζ, β.

Proof. The displayed definitions give $C_0 \leq S$, $R_0 \leq 4S^2$, $M \leq 17S^7$, and $T \leq 272S^{12}$. Substitute these into P_0 and use $d \geq 1$. For example, $\log(16(n+1)) \leq 4S$ for $S \geq 4$, which suffices for the bound on R_0; the remaining inequalities follow directly. □

6 From samples to an exact parallel circuit

Let $\mathcal{H}_P$ be the finite set of Boolean functions on X_d that admit globally unit-margin tanh representations of product at most P, allowing arbitrary finite depths and widths. It is finite because the domain is finite. Theorem 2.1 gives uniform circuit bounds for every member of this set.

Theorem 6.1 (Sample-to-parallel-circuit converse)**.** *Fix $d, n \geq 1$, $L \geq 3$, $\zeta > 0$, and $0 \leq \varepsilon < 1/2$. Use Theorem 5.1 with $\beta = 1/6$, and fix any one $N \geq N_0$ and any one $0 < \sigma^2 \leq N^{-(1+\zeta)}$. These choices are independent of the input law.*

Suppose that for every input distribution $\mathcal{D}$ on X_d, an iid n-sample labeled by f has positive probability of satisfying

$$\mathcal{D}\{x : p_S(f(x) \mid x) \leq 2/3\} \leq \varepsilon.$$

Put $\kappa = 1 - 2\varepsilon > 0$. Then f has a threshold circuit of size polynomial in $d + n + \kappa^{-1}$ and depth $O(\log(d + n + 2))$. It has a bounded-fan-in Boolean circuit of polynomial size in the same quantities and depth

$$O(\log^2(d + n + 2) + \log(\kappa^{-1} + 2)).$$

All polynomial degrees and depth constants are universal. They do not depend on the original teacher, L, N, or ζ.

Proof. Fix $\mathcal{D}$ and select a successful sample. The ceiling supplies a posterior event of mass at least $5/6$ on which the represented Boolean function belongs to $\mathcal{H}_{P_0}$. Condition on that event, obtaining a distribution ν on this finite class. On the good set

$$G = \{x : p_S(f(x) \mid x) > 2/3\},$$

whose $\mathcal{D}$-mass is at least $1 - \varepsilon$, a draw from ν has wrong-label probability strictly less than $(1/3)/(5/6) = 2/5$. Its signed vote therefore has expectation greater than $1/5$ at every $x \in G$.

Choose an odd integer $m_0 \geq 100(d + 1)$ no larger than $100(d + 1) + 2$. Independent draws of m_0 functions from ν have majority error at a fixed good input at most $e^{-m_0/50}$ by Hoeffding's inequality. A union bound over X_d is strictly less than one. Thus some majority g of m_0 members of $\mathcal{H}_{P_0}$ agrees with f on all of G, and has $\mathcal{D}$-error at most ε.

Let $\mathcal{K}$ be the finite class of all such m_0-voter majorities. This class is independent of $\mathcal{D}$. The conclusion just proved says

$$\text{for every } \mathcal{D}, \quad \max_{g \in \mathcal{K}} \mathbb{E}_{x \sim \mathcal{D}}[f(x)g(x)] \geq \kappa.$$

The finite minimax theorem, applied to the matrix with entries $f(x)g(x)$, gives a distribution λ on $\mathcal{K}$ with

$$f(x)\mathbb{E}_{g \sim \lambda} g(x) \geq \kappa \qquad \text{for every } x \in X_d.$$

Choose an odd integer $m_1 \geq 4(d + 1)/\kappa^2$ within two of this bound. Hoeffding and a union bound again give

$$2^d e^{-m_1 \kappa^2/2} < 1.$$

Some majority of m_1 members of $\mathcal{K}$ therefore equals f on every input. This is a majority of majorities of low-product functions; it is not being asserted to have low product itself.

Compile all its constituent functions by Theorem 2.1. Lemma 5.2 bounds P_0 by a universal polynomial in $d + n$. Put these circuits in parallel and add the two majority layers. Their number is polynomial in $d + n + \kappa^{-1}$, and the threshold depth increases only by two. For the Boolean circuit, convert the constituent circuits first, then implement the two majorities in depths $O(\log m_0)$ and $O(\log m_1)$ using the construction in Section 4. This gives the sharper additive $O(\log \kappa^{-1})$ term in the stated bound. □

Remark 6.2 (Quantifiers and mixtures). The hypothesis is distribution-free, and only existence of a successful sample is used for each distribution. The samples, low-product representatives, and both lists of voters may depend nonuniformly on f. The one width and one actual Gaussian scale in the theorem are fixed before the distribution is chosen. No posterior algorithm, posterior concentration on a single function, or pointwise statement about an individual posterior draw is assumed.

7 The critical endpoint: canonical features and parallel iteration

The critical prior has $t = 1$, equivalently $\sigma^2 = 1/N$. Its finite-width rate ceiling is different from the product ceiling above. We prove a separate circuit theorem for its near-baseline rate class. The proof preserves a finite set of anchor correlations, rather than the full feature covariance, and retains the learner depth in the resulting resource bound.

7.1 Variational paths and the imported scalar facts

For a finite input set $X \subseteq X_d$, put $Q_0(x, y) = (1 + x \cdot y)/d$. A critical controlled path with $H = L - 1$ hidden transitions consists of laws

$$\rho_j \ll \mathcal{N}(0, Q_{j-1}), \qquad Q_j(x, y) = \mathbb{E}_{\rho_j}[\tanh(G_x)\tanh(G_y)], \qquad D_j = \mathrm{KL}(\rho_j \| \mathcal{N}(0, Q_{j-1})).$$

Singular Gaussian references are allowed. A readout vector $z \in \mathrm{range}(Q_H)$ has cost

$$r = \tfrac{1}{2} z^T Q_H^\dagger z, \qquad A = \sum_{j=1}^{H} D_j + r.$$

Equivalently, if Q_H is the Gram matrix of feature vectors $u_H(x)$, then $z(x) = \langle c, u_H(x)\rangle$ for a vector c of squared norm $2r$. Let $J_L(f)$ be the infimum of A over full-cube paths with $f(x)z(x) \geq 1/2$ everywhere.

The scalar value $V(q)$ is the infimum of

$$\sum_{j=1}^{m} \mathrm{KL}(\eta_j \| \mathcal{N}(0, q_{j-1})) + \frac{1}{8q_m}, \qquad q_0 = q, \quad q_j = \mathbb{E}_{\eta_j} \tanh^2 G,$$

over all finite lengths, including length zero. We use the following previously proved facts from [2, "Scalar control and target-component constructions"]. They are stated explicitly to identify the only scalar-control input to our new parallelization argument.

Proposition 7.1 (Imported scalar Bellman facts)**.** *There is a limit $V_* := \lim_{q \downarrow 0} V(q) \in (1, 2)$ and universal $C_0, c > 0$ such that*

$$0 \leq V_* - V(q) \leq C_0 q \quad (0 < q \leq 2), \qquad V(q) \leq \frac{1}{8q}.$$

For a scalar transition from reference variance p to feature variance q, with entropy D, one has

$$D \geq V(p) - V(q), \qquad D \geq q - p + cp^2 \quad (0 < p \leq 2).$$

The constant c may be decreased so that $p - cp^2$ is increasing on $[0, 5]$.

The source states the small-state bound first and extends it by boundedness; increasing C_0 gives the interval above. Its strict potential proof gives dissipation on $[0, 1]$. The same proof extends to $[0, 2]$: for $q > 0$, $\log \mathbb{E} \exp\{\tanh^2(\sqrt{q}Z)\} < q$, its difference from q is $q^2 + O(q^3)$ at zero, and compactness supplies a positive uniform constant. The Bellman inequality is also immediate by appending a nearly optimal scalar suffix.

For $D \geq 0$, define $\mathcal{G}(D) \geq 1$ by

$$\mathcal{G}(D) - 1 - \log \mathcal{G}(D) = 2D.$$

Gaussian entropy duality gives, for any tilt ρ of a standard Gaussian with entropy at most D,

$$\mathbb{E}_\rho\langle g,u\rangle^2 \le \mathcal{G}(D)\,\|u\|_2^2, \qquad \sqrt{\mathcal{G}(D)} \le e^{\sqrt{D}}, \qquad \mathcal{G}(D) \le 4D+2. \tag{1}$$

Indeed, for a scalar second moment $m>1$, maximizing the entropy dual over quadratic functions gives $D \ge (m-1-\log m)/2$. For the second inequality write $v=\frac{1}{2}\log\mathcal{G}(D)$ and use $D=(e^{2v}-1-2v)/2 \ge v^2$. For the last, use $\log u \le u/2$ for $u \ge 2$. These facts also show that every linear coordinate lies in $L^2(\rho)$.

Lemma 7.2 (Variance and entropy occupation)**.** *Let a full-cube path have global margin* $1/2$ *and action* $A \le V_* + \epsilon$*, where* $\epsilon \ge 0$*. Put* $q_j = \max_x Q_j(x,x)$*. Then*

$$\sum_{j=0}^{H-1} q_j + \sum_{j=1}^{H}\sqrt{D_j} \le C\big[\log(H+2)+\sqrt{H\epsilon}\big]$$

for a universal C*.*

Proof. All one-input variances are positive: zero variance would propagate at finite entropy and make the final output zero. The initial diagonal is $(d+1)/d \le 2$, and subsequent diagonals are at most one. Project a transition onto an input attaining its new maximum, and use the monotonicity of $p-cp^2$ in Proposition 7.1. This gives

$$D_j \ge q_j - q_{j-1} + cq_{j-1}^2. \tag{2}$$

At an input attaining q_i, scalar projection of the suffix and Cauchy–Schwarz for the readout show that its remaining cost is at least $V(q_i)$. Therefore, with $S_i=\sum_{j\le i} D_j$,

$$S_i \le \epsilon + C_0 q_i, \qquad S_H \le V_* + \epsilon. \tag{3}$$

We give the occupation argument because both its conclusions will be used. On a consecutive interval of ℓ transitions with pre-states at least $u>0$, entropy expenditure E, and endpoint also at least u, dividing (2) by the squared pre-state and telescoping yields

$$\ell \le C(E/u^2 + 1/u). \tag{4}$$

The algebra is $(q'-q)/q^2 \ge 1/q - 1/q'$, with difference $(q'-q)^2/(q^2q') \ge 0$.

We also need a bound for an initial prefix of length t and entropy expenditure $E<3$:

$$\sum_{j<t} q_j \le C\log(t+2) + C\sqrt{tE}. \tag{5}$$

Let $F(v)=v-cv^2$. Compare q_j with r_j+s_j, where $r_0=q_0$, $s_0=0$, $r_{j+1}=F(r_j)$ and $s_{j+1}=F(s_j)+D_{j+1}$. These sequences are nonnegative, $r_j\le 2$, $s_j \le E<3$, and F is increasing on their range. Since $F(r+s)\le F(r)+F(s)$, induction gives $q_j \le r_j+s_j$. Also $r_j \le 1/(cj+1/r_0)$ and $c\sum_{j<t} s_j^2 \le E$. Summing and applying Cauchy–Schwarz proves (5).

If $\epsilon \ge 1/4$, summed dissipation and Cauchy–Schwarz give both claims directly from $S_H \le 2+\epsilon$. Otherwise $S_H<3$. Put $a=\max\{\epsilon,(H+1)^{-1}\}$ and $b=\min_j q_j$. If $a\ge 1/4$, then H is bounded; if $b\ge 1/4$, (4) bounds H. Thus assume $u_0=\max\{a,b\}<1/4$. Let $u_j=2^j u_0$, stop at the first $u_J \ge 1/4$, and define last-exit indices $\tau_j=\max\{i : q_i \le u_j\}$. There are $J=O(\log(H+2))$ bands.

For the initial prefix through τ_0, if $b\le a$ its entropy is $O(\epsilon+1/H)$ and its length at most H. Equation (5) gives its state sum, and Cauchy–Schwarz gives its square-root entropy sum, bounded

by $C[\log(H+2)+\sqrt{H\epsilon}]$. If $b > a$, its endpoint is b, its entropy is $O(b)$, and all its states are at least b. Equation (4) gives length $O(1/b)$; forced cooling bounds its state sum by $O(\log(H+2))$, and Cauchy–Schwarz bounds its square-root entropy by a constant.

For a nonempty later block $(\tau_j, \tau_{j+1}]$, its entropy is at most $\epsilon + 2C_0 u_j = O(u_j)$. After deleting its first transition, all pre-states and its endpoint exceed u_j, so its length is $O(1/u_j)$. Restoring the first transition preserves this bound. Its initial state is at most u_j, and each subsequent increase is bounded by the entropy spent; all states in the block are therefore $O(u_j)$. The block's state sum and square-root entropy sum are both $O(1)$. The tail after τ_J has bounded length and bounded contributions by the same occupation bound. Summing the blocks proves the lemma. □

7.2 Anchors and a finite canonical state

For two positive semidefinite kernels with identical anchor block $K_{TT} = K'_{TT}$, define

$$d_T(K, K') = \max_{x \in X_d} \sqrt{\|a_x - a'_x\|_2^2 + (\sqrt{v_x} - \sqrt{v'_x})^2},$$

where $a_x = K_{TT}^{\dagger/2} K_{Tx}$ and $v_x = K_{xx} - \|a_x\|_2^2$, and similarly for K'. This has a geometric meaning: identify the anchor spans isometrically, project each query onto that common span, and align its remaining orthogonal direction. It is also the root mean square difference in a Gaussian coupling that shares the anchor vector and the one-query residual Gaussian. Any coupling of query feature vectors with identical anchor vectors bounds this distance by its query root mean square error. No inverse Gram matrix will be evaluated by the final circuit.

Lemma 7.3 (Finite anchor replacement)**.** *Let a full-cube controlled path have action $A \le V_* + \epsilon$, and put $T_0 = 1 + \log(H+2) + \sqrt{H\epsilon}$. There is a genuine controlled path whose row densities depend only on a common set of*

$$q \le \mathrm{poly}(d, H, A+1) \exp(CT_0)$$

anchors, with the same anchor covariance blocks as the original path. It has an anchor-supported readout of global margin $1/2$ and action at most $A + C/H$. Its layer entropies are at most the original D_j. Its readout vector has norm at most $C\sqrt{A+1}$.

Proof. Write $I_j(T)$ for the entropy of the layer-j anchor marginal relative to its Gaussian reference. It is monotone in T and at most D_j. Choose $0 < \rho \le 1/64$ of order $[H(A+1)]^{-1}$, and a cutoff $\delta > 0$ specified below. Greedily add an input whenever either $I_j(T \cup \{x\}) - I_j(T) > \delta$ for some j, or projection of the final readout onto the anchor feature span changes its output at x by more than ρ. There are at most A/δ entropy additions. A readout addition increases the squared norm of its projection by at least ρ^2, since the final feature vectors have norm at most one. There are at most $2r/\rho^2$ such additions. This proves termination and a bound on the common anchor count.

At layer j replace the original density by its conditional expectation given G_T, denoted $w_j(G_T)$. This preserves the anchor marginal exactly and has entropy $I_j(T) \le D_j$. For a query x, the conditional entropy of G_x beyond G_T is at most δ. Conditional maximal coupling and Pinsker's inequality give a coupling of the original and replaced bounded tanh features, with identical anchor features, whose squared query error is at most $4\sqrt{\delta/2}$. Indeed the mismatch probability is conditional total variation, whose average is at most $\sqrt{\delta/2}$, and the squared feature difference is at most four. Thus the one-step d_T error is at most $C\delta^{1/4}$. Pinsker's inequality here follows from binary entropy after data processing onto a set attaining total variation.

Transfer the anchor density w_j to any preceding kernel with the same anchor block. Couple its Gaussian anchor-plus-query vectors as in the definition of d_T. The common density has entropy

at most D_j relative to this joint Gaussian law. Equation (1) and tanh's Lipschitz property imply that the resulting d_T error is at most $\sqrt{\mathcal{G}(D_j)}$ times the input error. Applying these replacements successively defines a genuine full-cube controlled path, with exact original anchor blocks. Its total query error is at most

$$CH\delta^{1/4}\exp\left(\sum_j \sqrt{D_j}\right).$$

Choose δ so this is at most $\rho/\sqrt{\max\{2r,1\}}$. Lemma 7.2 makes δ^{-1} at most a polynomial in $H, A+1$ times $\exp(CT_0)$, as needed for the anchor count.

The projected readout has norm at most $\sqrt{2r}$. Its query values change by at most ρ during the path transfer, in addition to the projection error ρ. Thus it has margin at least $1/2 - 2\rho$. Scale it by $(1-4\rho)^{-1}$ to restore margin $1/2$. This changes its action by $O(\rho A) = O(1/H)$ and preserves the asserted norm bound. The row entropy costs do not increase. □

Lemma 7.4 (Canonicalization with additive dimension)**.** *Let the query states $u_x \in \mathbb{R}^K$ include anchor states u_a. Let ρ be a tilt of $g \sim \mathcal{N}(0, I_K)$ of entropy at most D, and set $\Phi(u) = \tanh(\langle g, u\rangle)$ as a vector in $\mathcal{H} = L^2(\rho)$. There is a map $F : \mathbb{R}^K \to \mathbb{R}^{K+q+1}$, allowing unused zero coordinates, such that its Gram kernel has the same anchor block, all query-anchor inner products, and all query norms as Φ. Moreover*

$$F(u) = Au + B(u), \qquad \|A\|_{\mathrm{op}} \le \sqrt{\mathcal{G}(D)}, \qquad B(0) = 0,$$

and on the ball $\|u\| \le \sqrt{Q}$,

$$\mathrm{Lip}(B) \le C(Q + \sqrt{D}).$$

No assertion of full query-query kernel preservation is made.

Proof. Let $V \subseteq \mathcal{H}$ be the span of the K functions g_i and the q anchor functions $\Phi(u_a)$. For an orthonormal basis $e_1, \dots, e_M$ of V, with $M \le K + q$, define

$$F(u) = (\langle e_1, \Phi(u)\rangle, \dots, \langle e_M, \Phi(u)\rangle, \|(I - P_V)\Phi(u)\|).$$

Pythagoras preserves every norm. At an anchor the last coordinate is zero, which preserves every query-anchor inner product and the anchor block. Since $L(u) = \langle g, u\rangle \in V$, set

$$Au = (\langle e_b, L(u)\rangle)_{b=1}^M \oplus 0.$$

Equation (1) bounds its operator norm. Writing $R(u) = \Phi(u) - L(u)$, the residual is $B(u) = (P_V R(u), \|(I - P_V)R(u)\|)$ in these coordinates. Orthogonality and the reverse triangle inequality give

$$\|B(u) - B(v)\|^2 \le \|R(u) - R(v)\|^2_{L^2(\rho)}.$$

For the last bound put $X = \langle g, u\rangle$, $Y = \langle g, v\rangle$ and $Z = \langle g, u - v\rangle / \|u - v\|$ when $u \ne v$. The derivative of $\tanh s - s$ has absolute value $\tanh^2 s$, so

$$\frac{|R(u) - R(v)|^2}{\|u - v\|^2} \le W := Z^2 \min\{1, \max(|X|, |Y|)^4\}.$$

Under the reference Gaussian, Z is standard normal and $\mathrm{Var}(X), \mathrm{Var}(Y) \le Q$. For a fixed sufficiently small $c > 0$,

$$e^{cW} - 1 \le cZ^2(X^4 + Y^4)e^{cZ^2}.$$

Conditioning X or Y on Z bounds its fourth moment by $CQ^2(1+Z^4)$, regardless of their correlations. Hence $\log \mathbb{E}e^{cW} \le CQ^2$. Entropy duality gives $\mathbb{E}_\rho W \le C(D + Q^2)$, proving the Lipschitz assertion. □

Apply this canonicalization successively to the path from Lemma 7.3. At each stage the reference Gaussian can be written in the current finite coordinates, tilted by the prescribed anchor density. The next anchor-plus-one-query Gaussian law depends only on the three types of moments preserved by Lemma 7.4. Induction therefore gives $d_T = 0$ between the canonical recursion and the genuine anchor-only path. In particular their diagonals and anchored outputs are equal. After H transitions the canonical dimension is at most $d + 1 + H(q + 1)$. This does not claim that the canonical full covariance itself is the original controlled transition covariance.

7.3 Uniform finite evaluation of a canonical map

Lemma 7.5 (Polynomial canonical evaluators)**.** *In Lemma 7.4, suppose $K, q, \mathcal{G}(D), U, \eta^{-1}$ are bounded by a fixed polynomial in $R \geq 2$. On the ball $\|u\| \leq U$, the map F, and the residual $F(u) - Au$, admit uniform approximations of error η by polynomial-size constant-depth threshold circuits on a sufficiently fine inverse-polynomial input grid. The circuits use bounded dyadic constants of $O(\log R)$ bits. The polynomial bounds depend only on the stipulated resource polynomial, not on an anchor condition number or a description of ρ.*

Proof. Write

$$a(u) = \mathbb{E}_\rho[e(g)\tanh(g \cdot u)], \quad b(u) = \mathbb{E}_\rho \tanh^2(g \cdot u), \quad F(u) = (a(u), \sqrt{b(u) - \|a(u)\|^2}).$$

The radicand is nonnegative and $\|a(u)\| \leq 1$. By (1), $\mathbb{E}_\rho \|g\|^2 \leq K\mathcal{G}(D)$. Truncate to $\|g\| \leq B$. The discarded probability is at most $K\mathcal{G}(D)/B^2$. Bessel's inequality bounds the discarded vector expectation uniformly in u by the square root of that probability; the discarded scalar expectation is bounded by the probability itself. A polynomial B makes both errors any prescribed inverse-polynomial amount.

On the retained set use the importance density

$$p(g) = \frac{(1 + \|e(g)\|^2)\mathbf{1}_{\{\|g\| \leq B\}}}{Z}, \qquad Z \leq M + 1,$$

relative to ρ. The retained set has positive probability. Its vector and scalar integrands have respective bounds $\|e(g)\tanh(g \cdot u)/p(g)\| \leq Z/2$ and $\tanh^2(g \cdot u)/p(g) \leq Z$. They have Lipschitz bounds at most $ZB/2$ and $2ZB$. A net of the radius-U input ball at spacing proportional to $\theta/(ZB)$ has logarithmic cardinality $O(K \log(2 + UZB/\theta))$. The Hilbert-valued bounded-differences argument used in Lemma 3.4, and a union bound on this net, give a polynomial number of importance samples uniformly approximating a, b to error θ. Lipschitzness extends the bound to the whole ball.

The formulas have the form

$$\widehat{a}(u) = \sum_{\ell=1}^{m} c_\ell \tanh(g_\ell \cdot u), \qquad \widehat{b}(u) = \sum_{\ell=1}^{m} b_\ell \tanh^2(g_\ell \cdot u),$$

where $\|g_\ell\| \leq B$, $\|c_\ell\| \leq Z/(2m)$ and $0 \leq b_\ell \leq Z/m$. In particular, large coefficients in a possibly ill-conditioned formula for the basis e never appear in the evaluator. If both errors are at most $\theta \leq 1$, then

$$\left|\widehat{b} - \|\widehat{a}\|^2 - (b - \|a\|^2)\right| \leq 3\theta + \theta^2.$$

Clip the approximate radicand at zero. The inequality $|\sqrt{s} - \sqrt{t}| \leq \sqrt{|s - t|}$ shows that θ of order η^2 suffices for error η in F.

All incoming nodes and output coefficients are polynomially bounded and can be rounded on inverse-polynomial meshes. The matrix A is an independent nonuniform constant of bounded operator norm and can be rounded directly; it need not be estimated from truncated samples. Affine forms, tanh lookups, squares, finite sums, clipping, and square roots now have polynomial range and inverse-polynomial precision requirements. Use the constant-threshold-depth lookup and comparison construction from Section 4 for each. Square root needs the squared accuracy mesh. This proves the claim. It uses existence of finite quadrature constants, not an algorithm for the density or its orthonormal basis. □

Theorem 7.6 (Near-baseline critical rate implies parallel circuits)**.** *Suppose f has a globally $1/2$-margined critical path with H hidden transitions and action $A \le V_* + \epsilon$, $\epsilon \ge 0$. Put*

$$T_0 = 1 + \log(H+2) + \sqrt{H\epsilon}, \qquad R = (d + H + A + 2)e^{CT_0},$$

with a sufficiently large universal C. Then f has a threshold circuit of size $R^{O(1)}$ and depth $O(\log R)$, and a bounded-fan-in Boolean circuit of size $R^{O(1)}$ and depth $O(\log^2 R)$, with universal constants. Consequently, if $d + L \le Q^k$ and $J_L(f) \le V_* + k \log^2 Q/L$ for fixed k, then f has size polynomial in Q and threshold depth $O_k(\log Q)$.

Proof. Use Lemma 7.3 and the canonical recursion. Its dimension is polynomial in R. The genuine anchor-only path has action at most $A + C/H$, the same diagonals and readout as the canonical recursion, and row entropies at most D_j. Apply Lemma 7.2 to that genuine path for its diagonal sum, with the additional C/H absorbed in the constant, and to the original path for its original entropy sum. Together they give

$$\sum_j Q_{j-1}^{\max} + \sum_j \sqrt{D_j} \le CT_0.$$

Write its canonical maps as $F_j(u) = A_j u + B_j(u)$, with

$$\alpha_j = \sqrt{\mathcal{G}(D_j)}, \quad \|A_j\|_{\mathrm{op}} \le \alpha_j, \quad \mathrm{Lip}(B_j) \le \ell_j := C(Q_{j-1}^{\max} + \sqrt{D_j})$$

on the relevant state balls. Thus $\sum_j \ell_j \le CT_0$ and $\prod_j \alpha_j \le e^{CT_0}$.

Extend each B_j by radial projection onto its state ball. For finite evaluation, first floor each radius at R^{-c} and then round it upward on a dyadic mesh of spacing at most R^{-c}, for a sufficiently large universal c. The increase in the sum of squared radii is $O(HR^{-c}) \le O(1)$, so the total Lipschitz budget increases by at most a constant. The actual path is unchanged. These extensions fix zero, remain Lipschitz, and have polynomially bounded ranges. Radial projection is uniformly evaluable with polynomial resources by norm, square-root, and reciprocal lookups; division only occurs above the inverse-polynomial radius floor.

Let $S_j = \prod_{i \le j} \alpha_i$ and $v_j = u_j/S_j$, with $S_0 = 1$. The normalized recursion is

$$v_j = C_j v_{j-1} + H_j(v_{j-1}), \qquad C_j = A_j/\alpha_j, \quad \|C_j\|_{\mathrm{op}} \le 1,$$

where $H_j(v) = B_j(S_{j-1}v)/S_j$ uses the extended residual. Its Lipschitz constant is at most ℓ_j. All S_j lie between one and a polynomial in R.

Precompute the contraction products $C_{j:i} = C_j \cdots C_i$ as nonuniform matrices. Their entries have absolute value at most one and their dimensions are polynomial. The recursion is equivalent to the Volterra system

$$v_j = C_{j:1} v_0 + \sum_{i=1}^{j} C_{j:i+1} H_i(v_{i-1}),$$

where an empty product is the identity. Initialize $v_j^{(0)} = C_{j:1}v_0$ and compute all times in parallel by

$$v_j^{(k+1)} = C_{j:1}v_0 + \sum_{i=1}^{j} C_{j:i+1}H_i(v_{i-1}^{(k)}). \tag{6}$$

Put $\Lambda = \sum_i \ell_i = O(T_0)$. Expanding the error recurrence over strictly increasing time indices gives

$$\max_j \left\| v_j^{(k)} - v_j \right\| \le 2\sqrt{2}\,\frac{\Lambda^k}{k!}.$$

Indeed each product uses distinct ordered indices, and their sum is at most $\Lambda^k/k!$. Thus $k = O(\Lambda + \log(1/\eta)) = O(\log R)$ iterations suffice for any fixed inverse-polynomial error η.

Uniform additive error δ per update contributes at most

$$\delta \sum_{m\ge 0} \frac{\Lambda^m}{m!} = \delta e^{\Lambda} = \delta\, R^{O(1)}.$$

This treats finite quadrature and rounded arithmetic as errors relative to the exact Lipschitz maps; the approximating maps need not themselves have small Lipschitz constants. Clipped residuals are bounded, so the right side of (6) has polynomial norm at every iteration. This supplies a fixed polynomial evaluation range and a bootstrap for its inexact versions. Divide the primitive error tolerance by a sufficiently large polynomial to absorb all matrix, coefficient, and summation errors in δ.

Lemma 7.5 makes every simultaneous update a polynomial-size constant-depth threshold circuit. There are polynomially many time indices and matrix entries, and $O(\log R)$ iterations. The final rescaling by S_H and bounded readout amplify error only polynomially. Choose η, δ so that the final output changes by less than $1/16$; its $1/2$ margin then gives the exact signs. This proves the threshold bound. The majority-gate simulation in Section 4 gives the Boolean bound. For an infimum $J_L(f)$, choose a path within $1/H$ of that infimum. This adds only a constant to T_0, proving the final assertion without an attainment assumption. □

7.4 A finite-width full-cube rate ceiling

We next prove the extraction step needed to apply the structural theorem to the actual posterior, rather than a limiting rate model.

Lemma 7.7 (Finite-event covariance certificate)**.** *Consider the critical width-N prior on a set of m inputs, with H hidden layers, and put $k = m(m+1)/2$. For any measurable raw event E of probability at least e^{-NA}, $A \ge 0$, any $\tau > 0$, and $0 < \Delta \le 1$, there is a realization in E and laws $\rho_j \ll \mathcal{N}(0, Q_{j-1})$, relative to its empirical preceding covariances, such that*

$$\max_{x,y} \left| \mathbb{E}_{\rho_j} \tanh G_x \tanh G_y - Q_j(x,y) \right| \le \Delta, \qquad \sum_j \mathrm{KL}(\rho_j \| \mathcal{N}(0, Q_{j-1})) + r \le A + \tau,$$

provided

$$N \ge \frac{C}{\tau}\left[Hk \log\left(1 + \frac{C(A/\tau+1)Hk}{\Delta}\right) + m\log(C(A/\tau+1)) + 1 \right].$$

Here z is the selected realization's actual output and $r = \frac{1}{2} z^T Q_H^{\dagger} z$. No exchangeability of E is required.

Proof. For a row law μ on $h \in [-1,1]^m$, let $\phi(h)$ list its k upper-triangular products. Define the bounded dual

$$I_B(q;\mu) = \max_{\lambda \in [-B,B]^k} \{\lambda \cdot q - \log \mathbb{E}_\mu e^{\lambda \cdot \phi}\}.$$

A maximizer exists. Its exponential tilt ν has moments a, and coordinatewise optimality gives $\lambda_i(q_i - a_i) = B|q_i - a_i|$. Consequently

$$I_B(q;\mu) = \mathrm{KL}(\nu\|\mu) + B\,\|q-a\|_1 \geq 0. \tag{7}$$

The same feature tilt of the Gaussian preactivation has identical entropy and moments.

For N iid rows let $\widehat{q}$ be their empirical moments. A grid $\Gamma \subset [-B,B]^k$ of covering radius η has size at most $(1 + CB/\eta)^k$. The dual objective is $2k$-Lipschitz in λ, and each fixed exponential likelihood ratio has mean one. Hence

$$\mathbb{E} e^{N I_B(\widehat{q};\mu)} \leq e^{2Nk\eta}|\Gamma|.$$

This bound is uniform in the conditional reference row law. Iterate it forward through the network. Given the hidden layers, the readout has covariance Q_H/N, so, for $0 < s < 1$, $\mathbb{E}[e^{(1-s)Nr} \mid \text{hidden}] = s^{-\operatorname{rank}(Q_H)/2} \leq s^{-m/2}$. Writing I_j for the realized layer dual costs gives

$$\mathbb{E}_{\text{prior}} e^{N[\sum_j I_j + (1-s)r]} \leq e^{2NHk\eta}|\Gamma|^H s^{-m/2}.$$

Conditioning on E has relative entropy at most NA. Entropy duality therefore bounds the conditional expectation of $\sum I_j + (1-s)r$ by

$$A + 2Hk\eta + \frac{H \log|\Gamma| + (m/2)\log(1/s)}{N}.$$

Take $s = \tau/[16(A+\tau)]$, $\eta = \tau/(64Hk)$, and $B = (A+\tau)/\Delta$. Under the stated width bound some realization in E has the last cost at most $A + \tau/8$. Since all I_j are nonnegative and $(A+\tau)(1-s) = A + 15\tau/16$, its unweighted total $\sum I_j + r$ is at most $A + \tau$. Apply (7) layer by layer. Each moment ℓ_1 error is at most $I_j/B \leq \Delta$, proving the lemma. □

Lemma 7.8 (Consistent-path transfer)**.** *For a certificate as above with total cost at most B, there is a genuine controlled path of cost at most B, with output $\widehat{z}$, satisfying*

$$\max_x |\widehat{z}(x) - z(x)| \leq \sqrt{2B}\, H \sqrt{m\Delta}\, e^{\sqrt{HB}}.$$

Proof. Two positive semidefinite m-point kernels differing entrywise by at most Δ have feature embeddings with each corresponding row distance at most $\sqrt{m\Delta}$. To see this, their operator norm difference is at most $m\Delta$, and $\left\|K^{1/2} - K'^{1/2}\right\|_{\text{op}} \leq \sqrt{\|K - K'\|_{\text{op}}}$. The latter follows by monotonicity of the matrix square root from $K \leq K' + aI$ and $K' \leq K + aI$. Any existing embedding can be identified isometrically with its square-root embedding, enlarging the ambient Hilbert space when necessary.

Inductively couple Gaussian rows for the genuine preceding kernel and the empirical reference using such common feature embeddings. Tilt the joint Gaussian by the certificate's original density. The new marginal has no larger entropy by data processing, while (1) bounds each root mean square preactivation difference by $\sqrt{\mathcal{G}(D_j)}$ times its previous feature difference. Apply tanh, then use the certificate moment error to identify the controlled moment embedding with the next empirical

embedding at additional error $\sqrt{m\Delta}$. Couplings can be joined by identifying the common feature span and adjoining their orthogonal complements. Thus the maximal row error satisfies

$$e_j \le \sqrt{\mathcal{G}(D_j)}e_{j-1} + \sqrt{m\Delta}, \qquad e_0 = 0.$$

It follows that $e_H \le H\sqrt{m\Delta}\exp(\sum_j \sqrt{D_j}) \le H\sqrt{m\Delta}e^{\sqrt{HB}}$. Transfer the empirical readout vector of norm $\sqrt{2r}$ through this common embedding and project it onto the new final feature span. Its norm and readout cost do not increase, while its output error is at most $\sqrt{2r}\,e_H \le \sqrt{2B}\,e_H$. □

For the raw mass of a sample memorizer we use the established finite-width Fourier-seed result [2, "Finite-width target mass for Fourier seeds"]: there are universal constants such that a Boolean target g of Fourier variation $A_F(g)$ has raw critical probability at least

$$\exp\left[-N\left(V_* + C\frac{\log^2 R_g}{L}\right)\right], \qquad R_g = (d+L+2)A_F(g),$$

at every $N \ge R_g^C$ whenever $L \ge C\log R_g$. The event here has the original strict global margin and the signs of g. This is an imported finite-width target-mass theorem, not a rate-limit assertion.

Theorem 7.9 (Critical full-cube posterior ceiling)**.** *There are universal $C, c > 0$ such that the following holds. Put $R_0 = d + n + L + 2$, with $d, n \ge 1$, and suppose $L \ge C\log R_0$. At the actual critical scale $\sigma^2 = 1/N$, for every*

$$N \ge 2^{Cd}R_0^C + CL\log(1/\beta), \qquad 0 < \beta < 1,$$

and every consistently labeled sample of at most n inputs, the original posterior assigns probability at least $1-\beta$ to the fixed full-cube class

$$\mathcal{G} = \left\{g : J_L(g) \le V_* + c\frac{\log^2 R_0}{L}\right\}.$$

All its members have threshold circuits of polynomial size in R_0 and depth $O(\log R_0)$, uniformly over the class.

Proof. If $x_1, \dots, x_m$ are the distinct negative sample inputs, the function $g_S = 1 - 2\sum_{i=1}^m \mathbf{1}_{\{x_i\}}$ has the observed signs and Fourier variation at most $1 + 2n$. Indeed each point indicator has Fourier variation one. The imported target mass theorem gives a lower bound e^{-Na_0} for the conditioning denominator, where $a_0 = V_* + C_0\log^2 R_0/L$, at every $N \ge R_0^{C_0}$ after enlarging constants.

Let E be the raw event of globally margined draws whose Boolean function g has $J_L(g) > a_0 + K/L$, where K is a sufficiently large universal constant. It is measurable because there are finitely many sign functions. Suppose its probability were at least $e^{-N(a_0+1/L)}$. Apply Lemma 7.7 on the entire cube, with $m = 2^d$, $\tau = 1/L$, and action parameter $a_0 + 1/L$. Choose Δ sufficiently small that Lemma 7.8 gives output error at most $\rho = [CL(a_0+1)]^{-1} < 1/4$. It suffices to choose a tolerance with

$$\log(1/\Delta) \le C\big[d + \log(L(a_0+2)) + \sqrt{L(a_0+2)}\big].$$

The certificate's required width is thus at most $2^{Cd}R_0^C$. The transferred path preserves the selected realization's signs; multiplying its readout by $(1-2\rho)^{-1}$ restores margin $1/2$ and adds at most C/L action. It gives a path of action at most $a_0 + K'/L$ for one of the functions in E, contradicting $K > K'$.

Hence the bad raw mass is less than $e^{-N(a_0+1/L)}$. Divide by the sample-consistent denominator e^{-Na_0} to bound its posterior mass by $e^{-N/L} \le \beta$. Absorb K/L into the displayed logarithmic excess. Theorem 7.6 then gives the uniform circuit bounds. Its use of an arbitrarily near-minimizing path removes any issue of rate attainment. □

Corollary 7.10 (Critical sample-to-parallel-circuit converse)**.** *Let $0 \le \varepsilon < 1/2$, $\kappa = 1 - 2\varepsilon$, and $R_0 = d + n + L + 2$. Suppose $L \ge C \log R_0$ and choose one width above the threshold of Theorem 7.9 with $\beta = 1/6$. If, at $\sigma^2 = 1/N$, every input law gives positive probability to an n-sample with confidence-failure mass at most ε, then f has threshold size polynomial in $R_0 + \kappa^{-1}$ and depth $O(\log R_0)$. Its Boolean depth is $O(\log^2 R_0 + \log(\kappa^{-1} + 2))$, with polynomial size.*

Proof. Apply the two-majority argument of Theorem 6.1 with the fixed class $\mathcal{G}$ from Theorem 7.9 in place of $\mathcal{H}_{P_0}$. The one width is chosen before the input law, and the ceiling is simultaneous over every realized sample. Each voter now has the uniform circuit bound from Theorem 7.6. No other step changes. □

The exponential width threshold in this converse is deliberate: we extract on all 2^d inputs to obtain a distribution-independent full-cube circuit class. It does not assert a polynomial width threshold for this endpoint converse. It suffices for a necessary condition on a sample guarantee required at every admissible width.

8 Every fixed subcritical prefactor

Fix $0 < t < 1$ while allowing the width to grow. The corresponding exponent is $\gamma_N = 1 + 2\log(1/t)/\log N$, so a fixed-gap theorem does not apply uniformly in N. We prove a separate converse. All constants and polynomial exponents in this section may depend on the fixed value of t. Put

$$a = t^{-2} - 1, \quad c_t = \frac{a - \log(1+a)}{2}, \quad \delta_t = \frac{(a+1)\log(1+a) - (a+2)\log(1+a/2)}{2}.$$

A controlled path uses reference $\mathcal{N}(0, t^2 Q_{j-1})$, row entropies D_j, and readout action

$$r_t = \frac{1}{2t^2} z^T Q_H^{\dagger} z, \qquad A = \sum_{j=1}^{H} D_j + r_t, \qquad H = L - 1.$$

Its full-cube rate $J_{t,L}(f)$ is the infimum over such paths with signed output margin at least $1/2$.

8.1 A strict spectral covering gain

Lemma 8.1 (Entropy and two principal directions)**.** *Let $\rho \ll \mathcal{N}(0, t^2 I_K)$ have entropy at most D, where $K \ge 2$, and let $\lambda_1 \ge \lambda_2$ be the largest eigenvalues of $\Sigma = \mathbb{E}_\rho g g^T$. Set $\alpha = \sqrt{\lambda_1}$ and $b = (\lambda_1 + \lambda_2)/(2\sqrt{\lambda_1})$. Then*

$$\log \alpha \le (D - c_t)/a, \qquad \log b \le (D - c_t - \delta_t)/a, \qquad \delta_t > 0.$$

Every map satisfying $\|F(u) - F(v)\| \le \left\|\Sigma^{1/2}(u - v)\right\|$ sends any ball of radius r into two balls of radius rb.

Proof. Finite Gaussian entropy implies finite second moments, and absolute continuity makes Σ positive definite. Define $F_t(\lambda) = (\lambda/t^2 - 1 - \log(\lambda/t^2))/2$. The exact identity

$$\mathrm{KL}(\rho\|\mathcal{N}(0,t^2I)) = \mathrm{KL}(\rho\|\mathcal{N}(0,\Sigma)) + \sum_i F_t(\lambda_i)$$

uses the full second moment and allows a nonzero mean. Thus $F_t(\lambda_1) + F_t(\lambda_2) \le D$. The first gain bound follows from

$$F_t(\lambda) - c_t - \frac{a}{2}\log\lambda = \frac{a+1}{2}(\lambda - 1 - \log\lambda) \ge 0.$$

For the second put $s = \lambda_2/\lambda_1 \le 1$, $u = (a+1)(\lambda_1+\lambda_2)/(a+2)$ and $v = (a+2)/2$. Direct expansion gives

$$F_t(\lambda_1) + F_t(\lambda_2) - c_t - a\log b - \delta_t = v(u - 1 - \log u) + \log\frac{1+s}{2\sqrt{s}} + \frac{a}{2}\log\frac{2}{1+s} \ge 0.$$

The signs follow from $\log u \le u - 1$, arithmetic–geometric mean, and $s \le 1$. Also $\delta_t = 0$ at $a = 0$, and its derivative is $\frac{1}{2}\log((1+a)/(1+a/2)) > 0$ for $a > 0$.

For the covering statement, take two centers on the top principal axis of $\Sigma^{1/2}B_1$, at displacements $\pm(\lambda_1 - \lambda_2)/(2\sqrt{\lambda_1})$. At top input coordinate of magnitude $s \in [0,1]$, the squared distance to the nearer center is at most

$$\lambda_2 + (\lambda_1 - \lambda_2)(s^2 - s) + \frac{(\lambda_1 - \lambda_2)^2}{4\lambda_1} \le b^2.$$

The centers have preimages inside the input ball. Map these preimages through F, then translate and scale the ball. □

The canonical map from Lemma 7.4 has this metric property at every prefactor: the feature map $\Phi(u) = \tanh(\langle g, u\rangle)$ does in $L^2(\rho)$, and projection together with the residual norm is a contraction. All its anchor and diagonal identities remain valid. The gap $\delta_t \sim a^2/8$ as $t \to 1$ is positive only away from the endpoint.

8.2 Normalized anchors and finite evaluation

Lemma 8.2 (Normalized anchor replacement)**.** *Suppose a full-cube path has signed margin* $1/2$ *and* $A \le Hc_t + E$, $E \ge 0$. *Put* $R = (d + H + A + 2)e^E$. *There is a genuine path whose densities depend on a common set of* $R^{O_t(1)}$ *anchors, with exact original anchor blocks and an anchor-supported readout of margin* $1/2$ *and action at most* $A + C/H$. *Its row entropies are at most the original* D_j. *Its canonical state has dimension* $R^{O_t(1)}$.

Proof. Use the entropy upper gains

$$\alpha_j = t\sqrt{\mathcal{G}(D_j)}, \qquad S_j = \prod_{i\le j}\alpha_i, \quad S_0 = 1, \qquad \Gamma = S_H,$$

where $\mathcal{G}(D) - 1 - \log\mathcal{G}(D) = 2D$. They always use the *original* entropies, including after replacement. Every original or replacement row has second-moment operator norm at most α_j^2, so its feature norm is at most $\sqrt{2}S_j$. The tangent bound gives

$$\log\alpha_j \le (D_j - c_t)/a, \qquad \Gamma \le e^{E/a}.$$

Write $W = \sqrt{2t^2 r_t}$. The margin at any input implies $W\Gamma \geq 1/(2\sqrt{2})$, while $W \leq \sqrt{2t^2 A}$. The normalized readout multiplier and needed inverse margin tolerances are therefore polynomial in R.

Take $\rho_0 = [CH(A+1)]^{-1} \leq 1/64$. Select common anchors as in Lemma 7.3, with entropy cutoff η and readout projection tolerance ρ_0. There are at most $A/\eta + W^2/\rho_0^2$ anchors. Replace each density by its conditional expectation $w_j(G_T)$ given the anchors.

A one-dimensional Gaussian G of variance σ^2 satisfies

$$W_2^2(\nu, G) \leq 2\sigma^2 \,\mathrm{KL}(\nu \| G). \tag{8}$$

For a smooth increasing transport T from a centered Gaussian, change of variables and Gaussian integration by parts give

$$\mathrm{KL}(\nu \| G) = \frac{\mathbb{E}(T(X) - X)^2}{2\sigma^2} + \mathbb{E}[T'(X) - 1 - \log T'(X)].$$

This proves (8); translation and approximation give the general form. Conditional on the anchors, the reference query variance is at most $t^2 Q_{j-1}(x,x)$, and conditional entropy averaged under the tilted anchor law is at most η. Coupling original and replaced queries sharing that anchor vector gives feature root mean square error at most $\sqrt{2t^2 Q_{j-1}(x,x)\eta}$. After division by S_j this is at most $2(t/\alpha_j)\sqrt{\eta} \leq 2\sqrt{\eta}$. Zero conditional variance forces equality and needs no transport.

For transfer to a changed preceding kernel with the same anchor block, tilt the Gaussian query coupling by the common anchor density. Its entropy is at most D_j and its root mean square gain is at most α_j. Normalized propagation thus has gain at most one. Successive replacements define a genuine full-cube path of normalized query error at most $CH\sqrt{\eta}$. Choose η so $W\Gamma CH\sqrt{\eta} \leq \rho_0$. Its inverse and the anchor count are polynomial in R.

The anchored readout loses at most $2\rho_0$ margin. Scaling by $(1-4\rho_0)^{-1}$ restores margin $1/2$, adds $O(A\rho_0) = O(1/H)$ action, and gives new readout norm $W' \leq 2W$. Apply the canonical construction, retaining the span of the Gaussian coordinate functions and anchor features and appending the residual norm. It adds at most $|T|+1$ coordinates per layer. Anchor blocks, query-anchor moments, and diagonals agree inductively with the genuine replaced path. These are precisely the invariants used by subsequent anchored densities and the final readout; full query-query covariance preservation is not asserted. □

Lemma 8.3 (Evaluation after arbitrary scalar normalization)**.** *Let F be a canonical map on $\mathbb{R}^K$ for a controlled row law with second moment Σ and $M \leq K + q$ basis functions. For $\alpha \geq \sqrt{\|\Sigma\|_{\mathrm{op}}}$ and $S > 0$, the map $F(Sv)/(S\alpha)$ has a constant-threshold-depth, polynomial-size evaluator on $\|v\| \leq U$, to any inverse polynomial accuracy, whenever K, M, U and inverse accuracy are polynomially bounded. No bound on S, S^{-1} or the density is needed.*

Proof. Let $e(g) \in \mathbb{R}^M$ list an orthonormal basis of the canonical subspace, and put

$$k(g) = \|g\| / \alpha, \qquad \psi_g(s) = \frac{\tanh(S\,\|g\|\, s)}{S\,\|g\|}.$$

Assign zero at $g = 0$, a null event. Each ψ_g is zero-preserving and 1-Lipschitz, hence bounded by U on $[-U, U]$. The normalized feature is $k(g)\psi_g(\langle \widehat{g}, v\rangle)$, with $\widehat{g} = g/\|g\|$. Use the importance density

$$p(g) = \frac{1 + \|e(g)\|^2 + k(g)^2}{Z}, \qquad Z = 1 + M + \mathbb{E}k^2 \leq 1 + M + K.$$

The projection vector $b(v)$ and squared feature norm $q(v)$ are expectations, under this sampling law, of

$$\frac{e(g)k(g)}{p(g)}\psi_g(\langle \widehat{g}, v\rangle), \qquad \frac{k(g)^2}{p(g)}\psi_g(\langle \widehat{g}, v\rangle)^2.$$

Pointwise $\|ek/p\| \le Z/2$ and $k^2/p \le Z$. The vector summand is bounded by $ZU/2$ and has Lipschitz constant at most $Z/2$; the scalar summand is bounded by ZU^2 and has Lipschitz constant at most $2ZU$. Bounded concentration and a net of logarithmic cardinality $O(K \log(\mathrm{poly}(K, M, U)/\eta))$ give polynomially many nodes approximating both expectations uniformly to tolerance η. One may apply scalar concentration to each of the polynomially many vector coordinates. Complete the state by

$$(b(v), \sqrt{\max\{0, q(v) - \|b(v)\|^2\}}).$$

The square-root inequality only squares the required accuracy.

All sampled directions and coefficients have polynomial range. Replace each ψ_g by values on a polynomial mesh in $[-U, U]$. Its Lipschitz bound controls the error independently of $S\,\|g\|$. The bounded arithmetic and scalar lookup constructions proved earlier implement these sums, products, and the clipped square root at constant threshold depth. The physical scales never become circuit inputs. The construction is nonuniform. □

8.3 Structural and actual-posterior converses

Theorem 8.4 (Fixed-prefactor near-baseline parallelization)**.** *Fix $0 < t < 1$. If a full-cube path for f has signed margin $1/2$ and action $A \le Hc_t + E$, $E \ge 0$, then, with $R = (d + H + A + 2)e^E$, the target has threshold size $R^{O_t(1)}$ and depth $O_t(\log R)$. Its Boolean depth is $O_t(\log^2 R)$ with polynomial size in R.*

Proof. Use the anchored canonical path from Lemma 8.2, charge each row to its original D_j, and normalize by S_j from that proof. Each normalized map is globally 1-Lipschitz and zero-preserving. Normalized cube states therefore have norm at most $\sqrt{2}$ and polynomial dimension.

Let b_i be the two-ball factor of the actual canonical row covariance. Cover the first j layers, then propagate each ball through the remaining layers using the entropy upper gains α_i. The possible output error is at most

$$\sqrt{2}\,W' \prod_{i \le j} b_i \prod_{i > j} \alpha_i \le \sqrt{2}\,W' \exp\left(\frac{A + C/H - Hc_t - j\delta_t}{a}\right). \tag{9}$$

Every layer is charged exactly once. Choose $J = O_t(E + \log(A + 2) + 1)$ making this at most $1/64$. If $J \ge H$, retain all H layers. Otherwise there are at most 2^J prefix balls. Discard empty balls and replace each remaining center by an actual cube state in it, recording its label. This doubles the radius and gives stored normalized centers of norm at most $\sqrt{2}$. Intermediate existential cover centers need not be inside the cube-state ball; their images use the globally defined canonical maps.

The normalized suffix has output Lipschitz constant at most $W'\Gamma$, polynomial in R. Opposite labels at the retained layer are therefore separated by at least $1/(W'\Gamma)$, while every query is within $1/(32W'\Gamma)$ of a stored center. Declare it positive when some positive center is within $1/(4W'\Gamma)$. Inverse-polynomial state, center and squared-distance errors preserve this exact classifier.

Evaluate each retained normalized map by Lemma 8.3, projecting radially onto the radius-$\sqrt{2}$ ball after each approximate map. Projection is nonexpansive and fixes the true state, so errors add across the $O_t(\log R)$ maps. Its arithmetic uses a reciprocal only above the fixed positive radius threshold and has constant threshold depth at the required precision. Distance tests run in parallel.

Size is polynomial and threshold depth logarithmic; the previous Boolean majority simulation gives squared-logarithmic depth. □

We state the finite-width target-mass input explicitly. The companion scale-change construction [3, Proposition `prop:long` and the Fourier-seed section] shows that for $R_g = d + L + A_F(g) + 2$, $L \geq C \log R_g$, $N \geq R_g^C$, and $0 < t \leq 1$, the raw probability of the original strict global-margin event for g is at least

$$\exp\left[-N\left\{Hc_t + V_* + C\left(a \log R_g + \frac{\log^2 R_g}{H}\right)\right\}\right]. \tag{10}$$

Its constants are independent of t. This is an explicit finite-width target construction, not a rate-limit assertion. As at criticality, target mass is an analytic input; the extraction and parallelization arguments here are separate.

Theorem 8.5 (Fixed-prefactor full-cube posterior ceiling)**.** *Fix $0 < t < 1$, and let $R_0 = d+n+L+2$, $d, n \geq 1$ and $L \geq C \log R_0$. Define*

$$A_0 = Hc_t + V_* + C\left(a \log R_0 + \frac{\log^2 R_0}{H}\right), \quad H = L - 1.$$

There are constants C_t, K_t such that at every

$$N \geq 2^{C_t d} R_0^{C_t} + C_t H \log(1/\beta), \quad 0 < \beta < 1,$$

the original posterior at $\sigma^2 = t^2/N$ gives mass at least $1 - \beta$, simultaneously for every consistent sample of at most n points, to the sample-independent class

$$\mathcal{G}_t = \{g : J_{t,L}(g) \leq A_0 + K_t/H\}.$$

Every member has threshold size $R_0^{O_t(1)}$ and depth $O_t(\log R_0)$.

Proof. The sample memorizer from Theorem 7.9 has Fourier variation at most $1 + 2n$. Equation (10) bounds the conditioning denominator below by e^{-NA_0}.

Lemma 7.7 applies unchanged with references $\mathcal{N}(0, t^2 Q)$ and readout cost r_t. Hidden observables remain bounded; given them the readout has covariance $t^2 Q_H/N$ and $\mathbb{E}e^{(1-s)Nr_t} \leq s^{-m/2}$. Its width formula is unchanged. Consistent-path transfer now has recurrence $e_j \leq \alpha_j e_{j-1} + \sqrt{m\Delta}$ and final multiplier $\sqrt{2t^2 r_t}$. For total cost $B = O_t(H + \log R_0)$, each $\alpha_j = t\sqrt{\mathcal{G}(D_j)}$ is polynomial in $B + 1$. Suffix products may be larger, but their logarithms are at most $O_t(H \log(B+2))$. Hence a tolerance with

$$\log(1/\Delta) \leq C_t\{d + H \log(B+2) + \log(H(B+2))\}$$

makes the output error at most $\rho = [C_t H(B+1)]^{-1}$. The embedding and data-processing proof of Lemma 7.8 preserves the action. Readout scaling by $(1 - 2\rho)^{-1}$ restores margin $1/2$ and adds $O_t(1/H)$ action.

Consider the bad raw event of globally margined draws with rate above $A_0 + K_t/H$. If its probability were at least $e^{-N(A_0+1/H)}$, use the full-cube certificate with $m = 2^d$, $\tau = 1/H$ and the preceding tolerance. Its width requirement is at most $2^{C_t d} R_0^{C_t}$. The repaired path would give a bad function action at most $A_0 + K_t'/H$, contradicting $K_t > K_t'$. The bad raw mass is therefore less than $e^{-N(A_0+1/H)}$. Division by the denominator gives posterior bad mass at most $e^{-N/H} \leq \beta$.

Since $A_0 - Hc_t = O_t(\log R_0)$ under the depth condition, Theorem 8.4, applied within $1/H$ of each rate infimum, supplies uniform circuit bounds. No attainment assumption is needed. □

Corollary 8.6 (Fixed-prefactor sample-to-circuit converse)**.** *Fix $0 < t < 1$, $0 \le \varepsilon < 1/2$, and $\kappa = 1 - 2\varepsilon$. Let $L \ge C \log R_0$, $R_0 = d + n + L + 2$, and fix one width above the preceding ceiling with $\beta = 1/6$. If every input law gives positive probability to an n-sample with confidence-failure mass at most ε, at $\sigma^2 = t^2/N$, then f has threshold size polynomial in $R_0 + \kappa^{-1}$ and depth $O_t(\log R_0)$. Its Boolean depth is $O_t(\log^2 R_0 + \log(\kappa^{-1} + 2))$, with polynomial size.*

Proof. Use the two-majority and minimax proof of Theorem 6.1 with the fixed class $\mathcal{G}_t$. The width is chosen before the input law, the ceiling is simultaneous over all samples, and Theorem 8.4 supplies the uniform voter circuit bounds. □

Corollary 8.7 (Uniformity away from the critical prefactor)**.** *Fix $0 < t_0 < 1$. The structural theorem holds uniformly for $0 < t \le t_0$ with resource*

$$R = (d + H + t^2 A + 2) \exp(E/a).$$

The posterior ceiling and sample converse hold uniformly on this same range: their polynomial exponents and width constants depend only on t_0, if the ceiling class is replaced by

$$\mathcal{G}_t^{\mathrm{unif}} = \{g : J_{t,L}(g) \le A_0 + K/(t^2 H)\}.$$

In particular the finite-width threshold can be chosen as $2^{C_{t_0} d} R_0^{C_{t_0}} + C_{t_0} H \log(1/\beta)$, independently of t.

Proof. Put $Q = t^2 A$ and retain the entropy upper gains from Lemma 8.2. Use layer-dependent entropy cutoffs $\eta_j = \eta \mathcal{G}(D_j)$. Their total anchor count is at most

$$\sum_j \frac{D_j}{\eta \mathcal{G}(D_j)} \le \frac{H}{2\eta},$$

because $D/\mathcal{G}(D) \le 1/2$. The normalized conditional transport error is now $2(t/\alpha_j)\sqrt{\eta_j} = 2\sqrt{\eta}$. Take $\rho_0 = c/[H(1+Q)]$. Since $W^2 \le 2Q$ and $\Gamma \le e^{E/a}$, the required η^{-1} and readout projection count are polynomial in the displayed R, uniformly in t. Readout repair adds at most $CA\rho_0 \le C/(t^2 H)$ action and at most doubles its norm. The cover still charges the original hidden costs, so this norm factor is the only change needed in (9).

As a function of a, δ has $\delta(0) = 0$ and $\delta''(a) = 1/[2(a+1)(a+2)] > 0$. Thus $\delta(a)/a$ is increasing and bounded below by its positive value at $a_0 = t_0^{-2} - 1$. The mixed cover error is bounded by $\sqrt{2} W' \exp(E/a - j\delta_t/a)$, and $O_{t_0}(\log R)$ layers suffice. The arbitrary-scale evaluator requires no additional bound involving t^{-1}.

For the statistical assertion, the target budget satisfies

$$t^2 A_0 \le C(H + \log R_0 + 1), \qquad \frac{A_0 - H c_t}{a} \le C_{t_0} \log R_0.$$

These bounds persist after addition of $K/(t^2 H)$, since $(t^2 a)^{-1} = (1+a)/a \le 1 + a_0^{-1}$. In the finite-event certificate take $\tau = 1/(t^2 H)$ and event budget $B = A_0 + \tau$. Then $B/\tau = H t^2 B$ is polynomial in R_0 and $1/\tau = t^2 H \le H$. Transfer gains satisfy $\alpha_j^2 \le 4t^2 B + 2$, and the readout norm is at most $\sqrt{2t^2 B}$. With repair tolerance $\rho = c/[H(1 + t^2 B)]$, the required $\log(1/\Delta)$ has a polynomial bound independent of t. The certificate's width bound is therefore uniform, and readout repair adds $O(1/(t^2 H))$ action.

The bad-event contradiction now uses raw threshold $\exp[-N(A_0 + 1/(t^2 H))]$ and bad rate $A_0 + K/(t^2 H)$. Its posterior mass is at most $e^{-N/(t^2 H)} \le e^{-N/H}$. The structural resource of every retained function is polynomial in R_0, with constants depending only on t_0. The same two-majority argument proves the uniform sample converse. □

Thus arbitrarily small prefactors do not cause a loss of uniformity on a range bounded away from one. A prescribed prefactor may vary with dimension or width within that range. The direct spectral proof deteriorates at $t \to 1$; the next section resolves that boundary by combining block covering with critical parallelization.

9 Uniform parallelization across the critical boundary

We now remove the remaining dependence on a prefactor approaching one. The spectral argument alone would retain $O(S/a)$ layers and could create $2^{O(S/a)}$ representatives. Instead we cover blocks of $O(1/a)$ physical layers with a constant number of balls. A low-entropy prefix then has only $O(S)$ such blocks, and the critical parallel iteration evaluates the whole prefix.

Theorem 9.1 (Uniform near-baseline parallelization)**.** *Fix a numerical constant K_0. There are constants C_0, C, depending only on K_0, such that the following holds for every $0 < t \le 1$. Suppose a full-cube path for f has signed margin $1/2$ and*

$$A \le Hc_t + V_* + K_0\left(aS + \frac{S^2}{H}\right), \quad S \ge 1 + \log(d + H + 2), \quad H \ge C_0 S,$$

where $a = t^{-2} - 1$ and $c_t = (a - \log(1 + a))/2$. Then f has threshold size $\exp(CS)$ and depth CS. Its Boolean size is $\exp(CS)$ and depth at most CS^2. All constants are independent of t.

9.1 Adjusted costs and a controlled prefix

We first supply the scalar estimates used in its proof. The critical value V is nonincreasing: starting from a larger variance, use the same standardized law at each scalar step. This preserves its entropy cost and increases every subsequent variance by monotonicity of $\tanh^2$ under dilation. The terminal readout cost decreases. Taking infima proves the claim.

Put $q_j = \max_x Q_j(x, x)$, $q_0 = (d + 1)/d$, and

$$e_j = D_j - c_t - \frac{a}{2}\log(q_j/q_{j-1}), \qquad r_1 = t^2 r_t.$$

All query variances are positive, since zero variance would propagate at finite entropy and preclude a nonzero final output. At a query attaining q_j, let its preceding variance be $p \le q_{j-1}$ and put $M = \mathbb{E}G_x^2/p$. The scalar reference identity is

$$D_t(\eta) = D_1(\eta) + c_t + \frac{a}{2}(M - 1).$$

Since $q_j \le Mp$, the difference between e_j and this query's critical scalar entropy is nonnegative. Monotonicity of V and of $p - cp^2$ therefore gives

$$e_j \ge 0, \quad e_j \ge V(q_{j-1}) - V(q_j), \quad e_j \ge q_j - q_{j-1} + cq_{j-1}^2. \tag{11}$$

Moreover

$$\widetilde{A} := \sum_j e_j + r_1 = A - Hc_t - \frac{a}{2}\log(q_H/q_0) - ar_1 \le A - Hc_t + a.$$

The last inequality follows from $r_1 \ge 1/(8q_H)$ and $\frac{1}{2}\log(q_H/2) + 1/(8q_H) \ge -1$. Consequently, for $\epsilon = C_{K_0}(aS + S^2/H)$,

$$\widetilde{A} \le V_* + \epsilon, \qquad \sum_{j \le i} e_j \le \epsilon + Cq_i. \tag{12}$$

The latter follows by telescoping the Bellman inequalities over the suffix and using $r_1 \geq V(q_H)$.

Use the original entropy upper gains

$$\alpha_j = t\sqrt{\mathcal{G}(D_j)}, \quad P_j = \prod_{i \leq j} \alpha_i, \quad P_0 = 1, \quad \Gamma = P_H, \quad g_j = \log \alpha_j.$$

Their exact inverse relation gives

$$D_j - c_t = ag_j + \frac{1+a}{2}(e^{2g_j} - 1 - 2g_j), \qquad 0 \leq h_j := D_j - c_t - ag_j \leq e_j. \tag{13}$$

Here $q_j/q_{j-1} \leq \alpha_j^2$ proves the last inequality.

Lemma 9.2 (Uniform gain and last-exit bounds)**.** *Under the hypotheses of Theorem 9.1,* $\Gamma \leq e^{CS}$ *and* $t^2 A \leq CH$*. For* $0 < a \leq 1/4$*, the last index* τ *with* $q_\tau \leq a$ *satisfies*

$$H - \tau \leq C(\epsilon/a^2 + 1/a).$$

If no such index exists, the same bound holds for H*.*

Proof. Write $u_j = \log(q_j/q_{j-1})$. When $u_j > 0$, the critical entropy dominated by e_j and Gaussian second-moment duality imply $e_j \geq u_j^2/4$. Thus $u_j \leq 2\sqrt{e_j}$. If $g_j > \sqrt{e_j} \geq 0$, the nonnegative remainder in (13) is at least $(1+a)g_j^2$, contradicting $D_j - c_t = au_j/2 + e_j$. Hence $g_j \leq \sqrt{e_j}$. The critical occupation proof, using (11)–(12), gives

$$\sum_j g_j \leq C \log(H+2) + C\sqrt{H\epsilon}. \tag{*}$$

There is also the telescoping tangent inequality $g_j \leq u_j/2 + e_j/a$.

For clarity, its complementary last-exit estimate is

$$\sum_j g_j \leq C\{1 + \log(1/a) + \epsilon/a\}, \qquad 0 < a < 1/4, \quad \epsilon < 1/4. \tag{**}$$

To verify it, put $m = \min_i q_i$ and start dyadic thresholds at $b = \max(a, m)$. If $m \leq a$, the gain through the last state below a is at most $C + \epsilon/a$ by the tangent inequality and the prefix budget. If $m > a$, the reciprocal occupation bound proved in the critical section gives initial length at most $C(\epsilon/m^2 + 1/m)$ and cost at most $\epsilon + Cm$; Cauchy–Schwarz bounds its gain by $C(1 + \epsilon/m)$. Between the last exits below u and $2u$, remove one initial transition. Pre-states exceed u, the final state is at most $2u$, and the cost is at most $\epsilon + 2Cu$. Reciprocal occupation and Cauchy–Schwarz give length $C(\epsilon/u^2 + 1/u)$ and gain $C(1 + \epsilon/u)$. Sum geometrically up to a fixed positive threshold; the remaining tail has bounded length and gain. The same occupation calculation starting at threshold a proves the asserted last-exit length, without requiring small ϵ. If $m > a$, include the initial minimum-state prefix. This also covers the case with no state below a.

For the required gain, if $a \leq S/H$, use (*) to obtain CS. Otherwise $\epsilon/a \leq CS$ and $a \geq 1/H$. If $\epsilon \geq 1/4$ or $a \geq 1/4$, the whole-path tangent bound $(V_* + \epsilon)/a$ is $O(1 + \epsilon/a)$ in the respective case. In the remaining case use (**), with $\log(1/a) \leq \log H \leq S$. This proves the uniform gain. Finally $t^2 c_t \leq 1/2$, $t^2 a \leq 1$, and $S \leq H/C_0$ give $t^2 A \leq CH$. □

We prove Theorem 9.1. At $t = 1$ use Theorem 7.6; at $a \geq 1/4$ use Corollary 8.7, whose resource has logarithm $O(S)$. Hence suppose $0 < a < 1/4$.

Choose a large fixed K below and set $J = \lceil KS/a \rceil$. If $H \geq K_H S/a$, where K_H is sufficiently large compared with K and the last-exit constants, then $J \leq \tau$ and

$$E_{\mathrm{pre}} := \sum_{j \leq J} e_j \leq C_e a S,$$
$$D_{\mathrm{pre}} := \sum_{j \leq J} D_j \leq J c_t + C_e a S = O_K(aS), \qquad \sum_{j \leq J} \sqrt{D_j} = O_K(S). \tag{14}$$

The constant C_e is independent of K, since $S^2/H \leq aS/K_H$ and q_J is controlled by the last-exit prefix budget. More precisely, nonnegativity gives $E_{\mathrm{pre}} \leq \sum_{j \leq \tau} e_j \leq \epsilon + Ca$. Also $q_J \leq 1 \leq q_0$, so the telescoping logarithm in D_{pre} is nonpositive.

We need the sharper variance estimate

$$\sum_{j<J} q_j \leq C \log(J+2) + C\sqrt{J E_{\mathrm{pre}}} \leq C(1+\sqrt{K})S. \tag{15}$$

The forced-cooling comparison from the critical proof works for arbitrary total forcing by taking $r_{j+1} = r_j - c r_j^2$ and $s_{j+1} = \min\{1, s_j - c s_j^2 + e_{j+1}\}$. On a capped step $q_{j+1} \leq 1 \leq r_{j+1} + s_{j+1}$. On an uncapped step, monotonicity and subadditivity of $u - cu^2$ on $[0,3]$ preserve $q_j \leq r_j + s_j$. Capping preserves $c \sum s_j^2 \leq \sum e_j$. Now $\sum r_j = O(\log(J+2))$ and Cauchy–Schwarz give (15).

If instead $H < K_H S/a$, then $A \leq V_* + C_{K_H} S^2/H$. The full critical canonical/Picard proof applies to the raw subcritical costs with uniform constants. Indeed its scalar Bellman comparison is trivial when the next variance is no larger; when it is larger, $M > 1$ in the reference identity and $D_t \geq D_1$. Its raw dissipation is

$$D_t \geq q - t^2 p + c t^4 p^2 \geq q - p + c' p^2$$

since $t^2 \geq 4/5$. Gaussian gains are at most the critical gains, and the residual estimate after substituting $g = t\xi$ is at most $C(p + \sqrt{D_t})$. The readout cost also dominates the critical cost. Thus the occupation, anchor and Picard proof gives size $\exp(CS)$ and depth CS in this short regime. It remains to treat the long prefix above.

9.2 Gaussian and non-Gaussian row costs

Write $W = \sqrt{2t^2 r_t}$. Besides $W\Gamma \leq e^{CS}$, the absolute margin bounds every partial normalization:

$$P_j \leq 2W\Gamma \leq e^{CS} \qquad (j \geq 1). \tag{16}$$

Indeed $q_H(x) \leq q_j(x)(\Gamma/P_j)^2$, $q_j(x) \leq 1$, and $1/2 \leq W\sqrt{q_H(x)}$. Perform the adaptive normalized anchor replacement from Corollary 8.7, using the original α_j. Take readout tolerance $\rho = c/[H(1 + t^2 A)]$ and choose the entropy cutoff parameter η small enough that

$$W\Gamma C H\sqrt{\eta} \leq \rho, \qquad C H \sqrt{\eta} \max_j P_j \leq H^{-2}.$$

The inverse cutoff, anchor count, and canonical dimensions are all $\exp(O(S))$. The genuine replaced path has maximum diagonals $\widehat{q}_j$ satisfying

$$\sum_{j<J} \widehat{q}_j \leq \sum_{j<J} q_j + C. \tag{17}$$

This follows from $|\sqrt{\widehat{q}_j} - \sqrt{q_j}| \leq H^{-2}$ and bounded hidden variances. Its repaired readout has norm $W' \leq 2W$. The added action $O(1/(t^2 H))$ is absorbed by the theorem's budget after increasing K_0.

For any original controlled Gaussian row, decompose its entropy using its full second moment Σ:

$$D = J_{\mathrm{ng}} + C_{\mathrm{g}}, \qquad J_{\mathrm{ng}} = \mathrm{KL}(\rho \| \mathcal{N}(0, \Sigma)), \qquad C_{\mathrm{g}} = \sum_i F_t(\lambda_i(\Sigma)).$$

At the query attaining q_j, the Gaussian cost is at least $F_t(M)$, with $q_j/q_{j-1} \le M$. The tangent inequality for F_t implies $J_{\mathrm{ng},j}^{\mathrm{original}} \le e_j$. Anchor replacement does not increase this non-Gaussian entropy. To see this, split Gaussian coordinates orthogonally into the anchor span and its complement. The replaced law is its controlled anchor marginal times an independent free Gaussian. Its matching Gaussian has the same product form, so its J_{ng} equals that of the anchor marginal. Data processing of the original law and its matching Gaussian proves the inequality. Changed kernels with exact anchor block alter only the isometric realization and the free complement. The conclusion therefore persists in canonical coordinates. For the canonical row costs we have

$$J_{\mathrm{ng},j} + C_{\mathrm{g},j} \le D_j, \qquad \sum_{j \le J} J_{\mathrm{ng},j} \le C_e a S. \tag{18}$$

9.3 Blocks with a constant number of representatives

For a physical canonical map F_j, let A_j be its linear part, obtained from the retained Gaussian coordinate functions. Then $A_j^T A_j = \Sigma_j$. Its polar decomposition gives

$$A_j = t U_j (I + B_j), \qquad \|B_j\|_{\mathrm{HS}}^2 \le 2 C_{\mathrm{g},j},$$

where U_j is an isometry. The last inequality is $(\sqrt{\mu} - 1)^2 \le \mu - 1 - \log \mu$ summed over eigenvalues. For a block of b layers, telescoping yields

$$A_{\mathrm{block}} = t^b U_{\mathrm{block}} + E_{\mathrm{block}}, \qquad \|E_{\mathrm{block}}\|_{\mathrm{HS}} \le e^{\sqrt{2bC_{\mathrm{g,block}}}} \sqrt{2bC_{\mathrm{g,block}}}. \tag{19}$$

This holds also for increasing rectangular dimensions, since products of the U_j remain isometries.

The nonlinear residual $N_j = F_j - A_j$, on its physical state ball of squared radius $\widehat{q}_{j-1}$, satisfies

$$\mathrm{Lip}(N_j) \le C \sqrt{\lambda_j} (\lambda_j \widehat{q}_{j-1} + \sqrt{J_{\mathrm{ng},j}}), \qquad \lambda_j = \|\Sigma_j\|_{\mathrm{op}}. \tag{20}$$

Whiten by the matching Gaussian, $g = \Sigma_j^{1/2} \xi$. The law of ξ has entropy $J_{\mathrm{ng},j}$ relative to a standard Gaussian. The residual moment estimate in Lemma 7.4, followed by composition with $\Sigma_j^{1/2}$, proves (20). The canonical residual projection is a Hilbert-space contraction.

Set $b = \lceil c/a \rceil$, with a large fixed c, and call a full block Gaussian-good if $C_{\mathrm{g,block}} \le r_0 a$. Then $bC_{\mathrm{g,block}} \le (c + 1/4) r_0$, so (19) has a fixed bound M_0. Every subproduct of the actual linear gains has a fixed bound too, since $\sum_{\mathrm{block}} \sqrt{C_{\mathrm{g},j}} \le \sqrt{bC_{\mathrm{g,block}}}$. Telescoping $F_{\mathrm{block}} - A_{\mathrm{block}}$ on pairs of *actual query states* gives

$$\mathrm{Lip}(N_{\mathrm{block}}) \le C_{\mathrm{block}} \sum_{\mathrm{block}} (\widehat{q}_{j-1} + \sqrt{J_{\mathrm{ng},j}}). \tag{†}$$

All intermediate states for these pairs lie in the balls used in (20); both the full and linear maps have gain at most $\sqrt{\lambda_j}$. No ambient-ball assertion about this residual is needed.

Choose c so large that

$$\sigma_0 = e^{-c/(2(1+1/4))} < B_0/32, \qquad B_0 = e^{-(c+1/4)/4-1}.$$

Since $c_t \le a^2/4$, for every block cost $D_{\mathrm{block}} = \sum_{\mathrm{block}} D_j$,

$$t^b \le \sigma_0, \qquad \exp((D_{\mathrm{block}} - bc_t)/a - 1) \ge B_0.$$

In (19), truncate singular values of E_{block} below a sufficiently small fixed multiple of B_0. The retained rank is bounded by a constant depending only on c, r_0. A volumetric net for that fixed-rank ellipsoid therefore covers the image of any radius-r ball under A_{block} by at most M_{cover} balls of radius $B_0 r/8$, where M_{cover} is an absolute constant. The free isometry contributes at most $\sigma_0 r$, and the discarded operator and net errors each contribute a sufficiently small fixed multiple of $B_0 r$.

Also call a block nonlinear-good if

$$\sum_{\mathrm{block}} (\widehat{q}_{j-1} + \sqrt{J_{\mathrm{ng},j}}) \le \eta_{\mathrm{nl}},$$

where $\eta_{\mathrm{nl}} > 0$ is fixed so that (†) is at most $B_0/16$. In each nonempty linear cover cell select an actual query state. Two such states within the original cluster have distance at most $2r$, their linear images differ by at most $B_0 r/4$, and their nonlinear remainders differ by at most $B_0 r/8$. Thus the true outputs have an actual-representative cover of radius at most

$$\frac{B_0 r}{2} \le r \exp((D_{\mathrm{block}} - bc_t)/a - 1), \tag{21}$$

with at most M_{cover} cells. The selection of actual representatives is what permits the restricted estimate (†). On any bad block, simply propagate the existing center and use the global one-ball gain $\exp((D_{\mathrm{block}} - bc_t)/a)$. Treat the final incomplete block the same way.

There are $M = \lfloor J/b \rfloor$ full blocks, with $M \ge KS/[2(c + 1/4)]$ after increasing K. By (14), at most $(K/4 + C_e + 1)S/r_0$ are Gaussian-bad. Choose r_0 sufficiently large compared with c. Equations (15), (17) and (18) give

$$\sum_{j \le J} (\widehat{q}_{j-1} + \sqrt{J_{\mathrm{ng},j}}) \le C(1 + \sqrt{K})S.$$

After $c, r_0, \eta_{\mathrm{nl}}$ are fixed, choose K large enough that each type of bad block accounts for at most $M/4$. At least $KS/[4(c + 1/4)]$ blocks are good. Finally choose K_H to validate the long-prefix selection. This order of choices involves no problem parameter.

Start with an actual cube state and radius $2\sqrt{2}$. Multiplying good and bad block gains and dividing the prefix state by P_J gives normalized cover radius at most

$$2\sqrt{2} \exp\left(\frac{\sum_{j \le J} h_j}{a} - G_{\mathrm{good}} \right) \le 2\sqrt{2} \exp(C_e S - G_{\mathrm{good}}).$$

Enlarge fixed K so the subsequent readout multiplier $W'\Gamma \le e^{CS}$ leaves output uncertainty at most $1/64$. The number of stored representatives is $M_{\mathrm{cover}}^M = \exp(O(S))$, since $M = O(S)$. Every stored normalized center is an actual cube state, of norm at most $\sqrt{2}$. The normalized suffix has output Lipschitz constant at most $W'\Gamma$. Opposite labels are separated by $1/(W'\Gamma)$, so the same distance classifier as before is exact at inverse-exponential-in-S coordinate accuracy.

9.4 Parallel evaluation and the finite-width consequence

The prefix may contain $J \gg S$ layers, but (14) and (15) give

$$\sum_{j \le J} \sqrt{D_j} + \sum_{j < J} \widehat{q}_j = O_K(S).$$

The raw-reference residual estimate is $\mathrm{Lip}(F_j - A_j) \le C(\widehat{q}_{j-1} + \sqrt{D_j})$. After normalization by P_j, divide by $\alpha_j \ge t \ge \sqrt{4/5}$; its total Lipschitz budget is still $O(S)$ and the linear maps are contractions. Extend each residual by radial projection onto its actual normalized state ball. Floor that radius at $e^{-C_1 S}$. By (16), the total increase in physical squared radii is at most $He^{CS}e^{-2C_1 S}$, made at most one by fixed large C_1. This adds only a constant to the residual budget and leaves all actual states unchanged.

Precompute all interval products of the linear contractions and apply the strict-causal Picard iteration from Theorem 7.6. Its remainder is bounded by $C\Lambda^k/k!$, with $\Lambda = O(S)$, and uniform evaluation errors are amplified by at most e^{Λ}. Thus $O(S)$ iterations suffice at e^{-CS} primitive accuracy. All time indices and state dimensions are polynomial in e^S. Lemma 8.3 evaluates each normalized canonical map at constant threshold depth, independently of small P_j; subtracting its bounded linear part and projecting supplies the residual evaluator. Appending the parallel distance tests proves Theorem 9.1. □

Theorem 9.3 (Uniform actual-posterior circuit ceiling)**.** *There are universal positive constants*

$$C_{\mathrm{mass}}, \quad C_{\mathrm{pad}}, \quad C_{\mathrm{width}}, \quad C_{\mathrm{circ}}, \quad K$$

with the following property. Put $R_0 = d + n + L + 2$ and suppose $L \ge C_{\mathrm{pad}} \log R_0$. For every $0 < t \le 1$, define

$$A_0 = Hc_t + V_* + C_{\mathrm{mass}} \left(a \log R_0 + \frac{\log^2 R_0}{H} \right).$$

At every actual width

$$N \ge 2^{C_{\mathrm{width}} d} R_0^{C_{\mathrm{width}}} + C_{\mathrm{width}} H \log(1/\beta), \qquad 0 < \beta < 1,$$

and every consistent sample of at most n points, the original posterior at $\sigma^2 = t^2/N$ gives mass at least $1 - \beta$ to

$$\{g : J_{t,L}(g) \le A_0 + K/(t^2 H)\}.$$

Every such function has threshold size at most $R_0^{C_{\mathrm{circ}}}$ and depth at most $C_{\mathrm{circ}} \log R_0$. The mass-budget constant is fixed before choosing the padding, width, and circuit-resource constants; increasing a resource constant does not enlarge the defining rate class. Thus one common class of circuits with those resources is a posterior ceiling class for all t, N in the assertion.

Proof. Use the uniform target mass (10) for the sample memorizer. The finite-event argument of Corollary 8.7 was uniform in extraction, without requiring separation from $t = 1$: take normalized action $Q = t^2 A$, slack $\tau = 1/(t^2 H)$, and repair tolerance $c/[H(1 + Q)]$. The certificate ratios $A/\tau = HQ$ and $1/\tau = t^2 H \le H$, transfer gains $\alpha_j^2 \le 4Q + 2$, readout norm, and $\log(1/\Delta)$ all have uniform polynomial bounds. The raw-event contradiction gives exceptional posterior mass at most $e^{-N/(t^2 H)} \le e^{-N/H}$. The repaired rate threshold differs from A_0 by $O(1/(t^2 H))$, which is absorbed by $C(aS + S^2/H)$ for $S = C' \log R_0$. Theorem 9.1 now gives uniform circuit resources, including for paths approaching a rate infimum. The class of all circuits with these resource bounds is independent of t and of the sample. □

Corollary 9.4 (Uniform sample-to-parallel-circuit converse)**.** *Let $0 \le \varepsilon < 1/2$, $\kappa = 1 - 2\varepsilon$, $R_0 = d + n + L + 2$, and assume the padding hypothesis of Theorem 9.3. Fix one width above that theorem's threshold with $\beta = 1/6$, and any $0 < t \le 1$. If every input law has positive probability of an n-sample with confidence-failure mass at most ε, then f has threshold size polynomial in $R_0 + \kappa^{-1}$ and depth $O(\log R_0)$. Its Boolean depth is $O(\log^2 R_0 + \log(\kappa^{-1} + 2))$, with polynomial size. The constants do not depend on t.*

Proof. Apply the two-majority and minimax proof of Theorem 6.1 to the common circuit class from Theorem 9.3. Choose the one width and its actual prefactor before the input law. The ceiling is simultaneous over samples and its threshold is independent of the prefactor. □

10 A universal polynomial bound would imply a collapse

Lemma 10.1 (Bounded-weight tanh teachers for Boolean circuits)**.** *Every bounded-fan-in Boolean circuit of size s on d inputs has a tanh representation with global signed output margin greater than one, depth and width polynomial in $d+s$, and every physical weight an integer of absolute value at most 20. It has no biases after its augmented input.*

Proof. Use $\{-1,1\}$ wire values. AND, OR, NOT, and COPY have respective affine forms

$$u+v-1, \qquad u+v+1, \qquad -u, \qquad u,$$

of absolute margin at least one on Boolean inputs. Process gates in topological order, retaining all earlier values by copy gates; this gives width $O(d+s)$ and depth $O(d+s)$, with one initial layer if needed to encode input wires. Replace each gate by tanh of ten times its affine form. If every incoming value is within $1/100$ of its intended sign, the affine perturbation is at most $3/100$, and the new sign error is at most $2e^{-19.4} < 1/100$. Thus errors do not accumulate through the depth.

Implement each affine constant with a separate wire of intended value $+1$, initialized as $\tanh(10)$ from the augmented constant input and propagated by the same COPY map. Include its error in the preceding $3/100$ perturbation bound. Its gate coefficient is ± 10. Duplicate logical wire copies when needed so repeated gate inputs use separate coordinates; this changes width by only a constant factor. All hidden weights are then integers of absolute value at most ten. This also handles constant Boolean circuits. Finish with twice the encoded output as a linear readout. Its signed margin exceeds 1.98. Pad with copy layers if necessary to obtain weighted depth at least three.

For every larger common width N, pad all hidden layers by zero neurons and use zero readout weights on them. Since $\tanh(0) = 0$, this preserves the classifier, depth, and margin. On the finite cube, continuity gives an open parameter neighborhood with the same strict margin. Every nondegenerate Gaussian prior assigns that neighborhood positive probability, so these widths are realizable at every finite positive scale. □

Corollary 10.2 (Joint polynomial learning implies $\mathsf{P}/\mathrm{poly} = \mathsf{TC}^1$)**.** *Fix any prescribed prefactor schedule $0 < t = t(d,L,N) \le 1$, chosen independently of the input law and the observed sample. Suppose there is one polynomial sample bound jointly in $d, C, L, \varepsilon^{-1}, \log(2/\delta)$ with the following property. Every Boolean target admitting a depth-L, width-at-most-C tanh teacher with weights bounded by C and global signed margin greater than $1/2$ is learned by the original posterior at scale $\sigma^2 = t(d,L,N)^2/N$, for every input law and every integer $N \ge C$. The guarantee is that confidence-failure mass is at most ε with sample probability at least $1-\delta$. Then*

$$\boxed{\mathsf{P}/\mathrm{poly} = \mathsf{TC}^1.}$$

In particular $\mathsf{P}/\mathrm{poly} = \mathsf{NC}^2$. The same conclusion holds if the hypothesis instead covers every realizable width. In particular, the conclusion applies to every fixed exponent $\gamma \ge 1$, every fixed prefactor, and arbitrary prescribed prefactors approaching one.

Proof. Take any family in P/poly and a polynomial-size bounded-fan-in Boolean circuit family for it. Lemma 10.1 supplies teacher width, weight bound, and depth all polynomial in d. Enlarge C to bound both the width and twenty, and pad the teacher depth to at least d using the same robust copy gates. Fix, for example, $\varepsilon = 1/3$ and $\delta = 1/2$. The hypothesized sample count is $n = \mathrm{poly}(d)$, determined without choosing the learner width.

The padding $L \geq d$ and the polynomial bounds on n, L ensure $L \geq C_0 \log(d + n + L + 2)$ for all sufficiently large d. Choose one width above both C and the uniform ceiling threshold in Theorem 9.3 with $\beta = 1/6$. This threshold is independent of t, so the actual value $t(d, L, N)$ can now be used without circularly changing N. The width and scale have been fixed before the input law. Corollary 9.4 gives threshold size polynomial in d and depth $O(\log d)$, with uniform constants. No computability assumption on the prescribed schedule is needed for this nonuniform circuit conclusion. Finitely many remaining dimensions can be hardcoded. Hence $\mathsf{P}/\mathrm{poly} \subseteq \mathsf{TC}^1$. The reverse inclusion holds by definition. The Boolean simulation from Theorem 2.1 also gives $\mathsf{TC}^1 \subseteq \mathsf{NC}^2 \subseteq \mathsf{P}/\mathrm{poly}$. □

The width quantifier is essential here. A sample bound polynomial in the learner width N is not the joint bound in this corollary: it would make n depend on the width subsequently chosen above the ceiling threshold. Likewise, the result does not say that a fixed input distribution admitting polynomial sample complexity forces an exact parallel circuit.

Lemma 10.3 (Fixed learner depth, uniformly over the prefactor)**.** *For every fixed $L \geq 2$ there is a polynomial p_L such that every width-C, depth-L tanh teacher with weights bounded by C and global signed margin greater than $1/2$ is learned from*

$$n \geq \varepsilon^{-1} p_L(d, C, 1 + \log(1/\varepsilon), \log(2/\delta))$$

examples at every $N \geq C$ and every $0 < t \leq 1$. The polynomial is independent of N, t.

Proof. We give the target-cost bound, and state the additional intrinsic learning input precisely. Write $a_{N,t} = -N^{-1} \log \mu_{N,t}(A_f)$. The cloning cylinder of [4, Proposition `prop:cloning`] extends to all $0 < t \leq 1$ as follows. Triple the teacher readout and partition the N neurons into C groups of sizes k_b differing by at most one. The exact clone has first rows equal to the teacher rows, internal entries B^j_{ba}/k_a, and readout entries v_b/k_b. Constrain all scaled first-layer coordinates, the internal row projections onto the group-constant subspace, and all scaled readout coordinates to intervals of radius ρt about this clone; the respective scalings are $\sqrt{d}\, W^1$, $\sqrt{N}\, W^j$, and Nv. There are NM, where $M = d + 2 + (L-2)C$, independent constrained Gaussian coordinates of variance t^2, and their squared centers sum to at most NB_0, where

$$B_0 = d(d+1)C^2 + 2(L-2)C^4 + 36C^4.$$

The free internal matrices vanish on the group-constant subspace. An elementary sphere-net Gaussian estimate bounds all their operator norms by $b_L t$ with conditional probability at least $1/2$, where $b_L = \sqrt{16 \log 9 + 8 \log(4L)}$.

Put $A = 2 + 3C^2 + b_L$ and $\rho = [2 + 9C^2(d + L + 1)A^{L-2}]^{-1}$. The normalized hidden error obeys $e_1 \leq \rho(d+1)$ and $e_j \leq (\sqrt{2}C^2 + \sqrt{C} + b_L)e_{j-1} + \rho\sqrt{C}$. These inequalities hold for every $t \leq 1$ because both the cylinder errors and the free operator norms decrease with t. The output error is at most $3\sqrt{2}C^2 e_{L-1} + \rho \leq 1/2$, so the strict target margin is preserved. The scalar Gaussian interval estimate

$$\Pr\{|Z - u| \leq \rho t\} \geq \frac{2\rho}{\sqrt{2\pi}} \exp(-u^2/t^2 - \rho^2) \quad (Z \sim \mathcal{N}(0, t^2))$$

therefore gives

$$t^2 a_{N,t} \le B_0 + M \log \frac{e\sqrt{2\pi}}{2\rho} + \log 2 =: Q_L(d, C).$$

This is polynomial at each fixed L, uniformly in N, t.

The imported normalized intrinsic theorem [3, Proposition `prop:intrinsic`] gives a sample bound polynomial in $d, L, 1 + t^2 a_{N,t}$ and

$$\max\{1, t^H \mathcal{G}((a_{N,t} + t^{-2})/H)^{H/2}\}, \qquad H = L - 1,$$

apart from a factor $\varepsilon^{-1}(1 + \log(1/\varepsilon))^c$ and an additive $C\varepsilon^{-1}\log(2/\delta)$. It holds at every realizable width. Since $\mathcal{G}(D) \le 4D + 2$, the squared one-layer factor is at most $4(Q_L + 1)/H + 2$. Its $H/2$ power is polynomial for fixed L. This proves the claim, including the smaller widths handled by the intrinsic theorem's finite-width VC bound. □

Proposition 10.4 (A sufficient collapse)**.** *If* $\mathsf{P}/\mathrm{poly} = \mathsf{TC}^0$*, then there is one polynomial sample bound jointly in* $d, C, L, \varepsilon^{-1}, \log(2/\delta)$ *for the teachers in Corollary 10.2, simultaneously at every* $N \ge C$ *and every* $0 < t \le 1$*.*

Proof. Let $R = d + C + L + 2$. Lemma 11.2 below gives a Boolean circuit for each such teacher with at most R^q gates, where q is universal. Its finite description length is polynomial in R, even if the teacher's real weights have arbitrary precision.

Consider one universal Boolean circuit-evaluation family: given a circuit description and its input assignment, evaluate the circuit, and output zero on malformed descriptions. This family lies in P/poly, indeed in uniform polynomial time. Under the assumed equality it has one fixed majority depth h and one fixed polynomial size bound in the description length. Hardcode the teacher circuit description and leave its d Boolean inputs free. Layering and carrying constants changes h by only a fixed factor. This produces a threshold circuit of depth at most h with width and coefficient magnitudes polynomial in R, and normalized gate margins at least one. The usual integer form of majority gates has all these properties. Crucially, the same depth and polynomial degree work for every teacher because the assumption was applied to one universal family.

For $L \ge h + 1$, apply [3, Theorem `thm:threshold`], including its short-padding case. Its resource exponent $C(h + 1)^2$ is fixed, so its bound is jointly polynomial in the stated resources and statistics. It holds at every realizable width; each $N \ge C$ is realizable by padding the original teacher. In particular, the learner width need not exceed the compiled circuit width. For the finitely many depths $2 \le L < h + 1$, apply Lemma 10.3 and take the largest of their polynomial degrees and constants. □

Thus the two complexity assumptions bracket the desired learning statement:

$$\mathsf{P}/\mathrm{poly} = \mathsf{TC}^0 \implies \text{universal joint-polynomial learning} \implies \mathsf{P}/\mathrm{poly} = \mathsf{TC}^1.$$

This does not identify either implication as an equivalence. The sufficiency proof needs no efficient construction of the compiled circuits or computation of the posterior.

11 An exponential depth lower bound would separate circuit classes

We first record two elementary facts needed to control the resource and subsequence quantifiers. The first uses only exact sample consistency, and holds for any posterior of this kind.

Lemma 11.1 (Finite-domain occupancy bound)**.** *Let X be a finite domain of size M. A posterior supported on classifiers agreeing with all observed labels has, for every input law, confidence-failure mass at most ε with probability at least $1-\delta$ whenever*

$$n \geq \frac{M \log 2 + \log(1/\delta)}{\varepsilon}.$$

Proof. At every observed input the posterior correct-label probability is one. The confidence-failure set is therefore contained in the unseen set U. Any fixed subset of mass greater than ε is missed by all samples with probability at most $e^{-n\varepsilon}$. A union bound over the 2^M subsets gives $\mathbb{P}\{\mathcal{D}(U) > \varepsilon\} \leq 2^M e^{-n\varepsilon}$. □

Lemma 11.2 (Margined tanh teachers have polynomial-bit descriptions)**.** *Suppose a tanh teacher has depth D, hidden widths at most C, weights bounded in magnitude by $B \geq 1$, and signed output margin greater than $1/2$ on X_d. Its Boolean function has a Boolean circuit of size polynomial in*

$$d + C + D + \log(2 + B).$$

No computability or precision restriction on the original real weights is required.

Proof. Put $m = \max\{d+1, C, 2\}$, $b = \lceil \log_2(B+2) \rceil$, $q = \lceil \log_2 m \rceil$, and

$$p = (D+3)(b+q+4) + 10, \qquad \rho = 2^{-p}, \qquad A = 4m(B+1).$$

Round each weight to a multiple of ρ within error ρ. Evaluate each affine form exactly on dyadic hidden states, and approximate each hidden tanh to error at most ρ, with its output clipped to $[-1,1]$. If e_j is the largest coordinate error at layer j, the Lipschitz property gives

$$e_j \leq mBe_{j-1} + (m+1)\rho \leq Ae_{j-1} + A\rho.$$

The same bound covers the final linear layer. Since $\log_2 A \leq b+q+2$, we have $e_D \leq 2\rho A^D < 1/16$, preserving every output sign. The rounded parameters have $O(b+p)$ bits, and the exact affine forms have $O(b+q+p)$ integer and fractional bits. Their denominator is 2^{2p}.

The activation evaluator must handle large preactivations without enumerating a range of size B. First inspect the sign and compare $|u|$ to $p+10$. If $|u| \geq p+10$, saturation to the appropriate sign has error at most $2e^{-2(p+10)} < \rho$. Otherwise use oddness and assume $0 \leq u \leq p+10$. Set $J = \lceil 16(p+10)^2 \rceil$ and evaluate

$$E_J = \sum_{k=0}^{J} \frac{(2u)^k}{k!}.$$

The exponential-series remainder is at most

$$e^{2u}\left(\frac{2eu}{J+1}\right)^{J+1} < 2^{-p-4}.$$

Compute $(E_J - 1)/(E_J + 1)$, and round to the nearest ρ-grid point, clipped to $[-1,1]$. Since $E_J \geq 1$, the derivative of this quotient in E_J is at most $1/2$; the total activation error is therefore less than ρ.

The Taylor terms have common denominator $2^{2pJ}J!$. Its bit length and those of the numerators are $O(p^3 + p^2 \log p)$. Ordinary integer arithmetic thus evaluates the series, division, and rounding with polynomially many Boolean gates. There are only polynomially many affine operations and activation calls in the teacher resources. Since $p = O(D(\log(m+1) + \log(B+2)))$, the claimed circuit-size bound follows. The approximation algorithm is uniform on its rounded parameter strings, even though those strings may be chosen nonuniformly from the original real weights. □

The second imported input is a weaker common form of the previously proved circuit-target learning bounds. We restrict its statistical parameters to fixed constants because that is all the separation argument needs.

Proposition 11.3 (Imported circuit-target upper bound)**.** *Fix $0 < \varepsilon, \delta < 1$. In either of the following regimes, there is a constant $a \geq 1$ such that the assertion below holds:*

(i) a fixed $\zeta > 0$, all scales $0 < \sigma^2 \leq N^{-(1+\zeta)}$, with a allowed to depend on $\zeta, \varepsilon, \delta$;

(ii) the critical scale $\sigma^2 = N^{-1}$, with a allowed to depend on ε, δ.

If a target has a bounded-fan-in Boolean circuit of wire size s and depth $h \geq 1$, then at any learner depth satisfying

$$L \geq a(h+1)\log(d+s+L+2),$$

a sufficient sample count for the original posterior is

$$n \leq (d+s+L+2)^{a(h+1)}.$$

This guarantee holds for every input law and every $N \geq C_L(f)$, where $C_L(f)$ is the least width admitting a depth-L tanh representation with weights bounded by that width and signed output greater than $1/2$ everywhere. In regime (i) it is simultaneous over the displayed finite scales. In regime (ii) it holds, more generally, at every realizable width.

This is a direct weakening of [1, "Precision-independent threshold-circuit learning"] in regime (i), and [2, "Real threshold teachers without a weight bound"] in regime (ii). To match their conventions, layer a bounded-fan-in circuit by inserting copy wires; its size and width grow by at most a polynomial factor, and its depth stays $O(h)$. The fixed statistical factors and this polynomial change are absorbed by a. The source in regime (i) in fact needs only $L \geq 2h$ before this harmless weakening. A realizable width is one at which a depth-L tanh network realizes the target signs; scaling the readout then gives strict margin $1/2$ and positive Gaussian target mass. In particular, every width at least that of a specified depth-L teacher is realizable. The bounded-weight admissibility condition $N \geq C_L(f)$ can be stronger than realizability; we keep the two notions distinct. No new Gaussian target-mass claim is needed for this proposition.

Theorem 11.4 (Exponential depth hardness implies $\mathsf{NC} \neq \mathsf{P/poly}$)**.** *Fix either regime of Proposition 11.3, and fixed $0 < \varepsilon, \delta < 1$. Suppose there is a sequence of targets $f_j : X_{d_j} \to \{-1, 1\}$ with tanh teachers at the same learner depths $L_j \to \infty$, widths at most C_j, weights bounded by C_j, and global signed margins greater than $1/2$. Assume*

$$d_j + C_j \leq L_j^b$$

for some fixed b, and that obtaining the distribution-free, all-admissible-width confidence guarantee, over $N \geq C_{L_j}(f_j)$, requires at least $\exp(cL_j)$ examples along this sequence, for some fixed $c > 0$. The same conclusion applies to the weaker width range $N \geq C_j$. Then

$$\boxed{\mathsf{NC} \neq \mathsf{P/poly}.}$$

Indeed a subsequence of the targets extends to a Boolean family in $\mathsf{P/poly} \setminus \mathsf{NC}$.

Proof. Lemma 11.1, with $M = 2^{d_j}$, gives a universal $O(2^{d_j})$ sample upper bound for fixed ε, δ. The assumed lower bound therefore forces

$$d_j \geq \frac{c}{\log 2} L_j - O(1).$$

Consequently $L_j = O(d_j)$ and $C_j \leq \mathrm{poly}(d_j)$, while also $d_j \leq \mathrm{poly}(L_j)$. Select an infinite subsequence with distinct strictly increasing d_j. Define a Boolean family F_d to equal f_j at those dimensions and a constant at every other dimension. Lemma 11.2 puts this family in P/poly. Nonuniformity permits both the arbitrary subsequence and the rounded teacher coefficients; no effective selection is being asserted.

Suppose $F \in \mathsf{NC}$. Then, for some fixed k and fixed size exponent, the selected targets have circuits of size $s_j = \mathrm{poly}(d_j)$ and depth $h_j = O(\log^k(d_j + 2))$. All of d_j, s_j are polynomial in L_j. Hence the depth condition in Proposition 11.3 holds for all sufficiently large j, and that proposition supplies a sample upper bound

$$n_j \leq \exp(O(\log^{k+1}(L_j + 2))) = \exp(o(L_j)).$$

It holds at every admissible width $N \geq C_{L_j}(f_j)$, in particular throughout $N \geq C_j$, and in regime (i) at every displayed scale. This contradicts the exponential lower bound. Thus $F \notin \mathsf{NC}$. □

The separation implication applies to nonexplicit targets as well as explicit ones. A counting or diagonalization construction of targets with the stated polynomial teacher resources would still produce the separated family. The global margin is important: it is what allows arbitrary original real coefficients to be replaced by finite descriptions.

Corollary 11.5 (Every fixed stretched-exponential rate has the same barrier)**.** *Under the teacher, statistical, and width hypotheses of Theorem 11.4, replace the sample lower bound* $\exp(cL_j)$ *by* $\exp(cL_j^\alpha)$*, for any fixed* $c, \alpha > 0$*. Then the same conclusion* $\mathsf{NC} \neq \mathsf{P}/\mathrm{poly}$ *holds. In particular this applies at every fixed Gaussian exponent* $\gamma \geq 1$*.*

Proof. The occupancy bound now forces $d_j \geq (c/\log 2) L_j^\alpha - O(1)$. Hence $L_j = O(d_j^{1/\alpha})$, and both depth and teacher width are polynomial in d_j. The finite-description and subsequence argument still give a family in P/poly. If it belonged to some NC^k, the imported learning theorem would give $\exp(O(\log^{k+1}(L_j + 2))) = \exp(o(L_j^\alpha))$ examples, contradicting the assumed lower bound. □

Thus a fixed stretched-exponential rate is already stronger than merely disproving a polynomial sample bound. The conditional contrapositive of Corollary 10.2, under $\mathsf{P}/\mathrm{poly} \neq \mathsf{TC}^1$, supplies failure of a universal polynomial guarantee; it supplies no stretched-exponential rate. Such a rate would also entail the stronger separation in Corollary 11.5.

Corollary 11.6 (The lower-bound barrier is uniform over all prefactors)**.** *The conclusion of Corollary 11.5 also holds for an arbitrary sequence of actual prefactors* $0 < t_j \leq 1$ *and realizable learner widths, with* $\sigma_j^2 = t_j^2/N_j$*. No separation from* $t = 1$ *is required.*

Proof. Here we use the companion uniform positive theorem [3, Theorem `thm:threshold`, long-padding case] in place of Proposition 11.3. Its precise specialization needed here is: for fixed statistical parameters, a bounded-fan-in Boolean circuit of size s and depth h has sample bound

$$(d + s + L + 2)^{C(h+1)}$$

at every realizable width and every $0 < t \leq 1$, whenever $L \geq C(h+1)\log(d + s + L + 2)$. Boolean gate coefficients and margins are absolute constants; layering changes only the universal polynomial

factors. The occupancy and finite-description parts of the preceding proof do not depend on the scale. If the resulting family were in NC^k, this uniform theorem would give $\exp(O(\log^{k+1} L_j))$ examples at the very same actual N_j, t_j, contradicting $\exp(cL_j^\alpha)$. □

Corollary 11.7 (Polynomial coefficient bit length suffices for the barrier)**.** *Fix $c, \alpha, b > 0$ and fixed accuracy and confidence parameters in $(0, 1)$. Suppose targets at learner depths $L_j \to \infty$ have tanh teachers at those same depths, with widths C_j and weight magnitudes at most $B_j \geq 1$, and signed output margins greater than $1/2$, with*

$$d_j + C_j + \log(2 + B_j) \leq L_j^b.$$

If the original posterior requires $\exp(cL_j^\alpha)$ examples for the fixed distribution-free confidence guarantee at some actual realizable widths N_j and prefactors $0 < t_j \leq 1$, then $\mathsf{NC} \neq \mathsf{P/poly}$. In particular the same conclusion applies to a lower bound on a guarantee required at every realizable width.

Proof. The occupancy bound gives $d_j = \Omega(L_j^\alpha)$, so the teacher width, depth, and coefficient bit length are polynomial in d_j. Lemma 11.2 therefore places a subsequence of these targets in a family in $\mathsf{P/poly}$. If that family belonged to some fixed NC^k, its targets would have Boolean circuits of size polynomial in d_j and depth $O(\log^k d_j)$. Since also $d_j \leq L_j^b$, the uniform target theorem used in Corollary 11.6 applies at the same learner depth, width, and scale and gives $\exp(O(\log^{k+1} L_j)) = \exp(o(L_j^\alpha))$ samples, a contradiction. The all-realizable-width version of that upper theorem needs no condition $N_j \geq B_j$. □

Thus polynomially many relevant coefficient bits, rather than polynomial coefficient magnitude, suffice for the lower-bound barrier. This does not cover an activation that itself encodes hard truth tables: the proof uses the polynomial-time numerical evaluator for exact tanh. Large but polynomial-bit coefficients do not substitute for such an activation-level lookup mechanism.

Corollary 11.8 (A quantitative threshold-depth obstruction)**.** *Under the hypotheses of Corollary 11.7, suppose $0 < \alpha \leq 1$. For every fixed $a > 0$, there is a constant $c_a > 0$ such that any real threshold circuit of width at most L_j^a computing f_j has depth at least*

$$c_a \frac{L_j^\alpha}{\log(L_j + 2)}$$

for all sufficiently large j. Its original coefficients may be arbitrary real numbers. The constant may depend on the fixed resource exponents and statistical parameters.

Proof. It suffices to consider threshold depth $h_j \leq L_j$. The weight-independent threshold compiler and the uniform target upper bound give sufficient sample count

$$\exp(C(h_j + 1) \log R_j), \qquad R_j = d_j + L_j^a + h_j + L_j + 2,$$

provided $L_j \geq C(h_j + 1) \log R_j$; universal polynomial compiler overhead changes only C. Write $\log R_j \leq B_a \log(L_j + 2)$. If

$$h_j + 1 \leq \kappa L_j^\alpha / \log(L_j + 2), \qquad \kappa \leq \frac{\min\{1, c\}}{2CB_a},$$

then the padding condition holds because $L_j^\alpha \leq L_j$, and the sample upper is at most $\exp(cL_j^\alpha/2)$. This contradicts the assumed lower bound at the same width and scale. Absorb the additive one in $h_j + 1$ by decreasing the constant, since $L_j^\alpha / \log(L_j + 2) \to \infty$. □

12 Scope and remaining questions

The product structural theorem is unconditional and applies to arbitrary finite depths, widths, and neuronwise zero-preserving 1-Lipschitz activations. Its statistical consequence uses the exact original tanh posterior at finite Gaussian scale. The product route covers every fixed gap $\gamma \geq 1 + \zeta$ without a resource dependence on the learner depth in the circuit conclusion. Canonical features, near-baseline action, and the block cover give a complementary converse uniform over all $0 < t \leq 1$, equivalently all $0 < \sigma^2 \leq 1/N$, with the depth padding, depth-dependent circuit resources, and finite-width threshold in Corollary 9.4. This includes arbitrary prescribed schedules approaching the critical endpoint and all fixed exponents $\gamma \geq 1$. The actual width is chosen before the input law, and no limiting posterior is substituted for the original one. Corollary 11.6, using the companion uniform positive theorem, also makes the stretched-exponential lower-bound barrier uniform throughout this scale range.

The two complexity consequences address different strengths of learning claim. A universal joint polynomial bound would imply $\mathsf{P/poly} = \mathsf{TC}^1$, while the stronger assumption $\mathsf{P/poly} = \mathsf{TC}^0$ suffices for it. Even a fixed stretched-exponential lower bound $\exp(\Omega(L^\alpha))$, $\alpha > 0$, with polynomial teacher resources would imply $\mathsf{P/poly} \neq \mathsf{NC}$. Neither implication is an equivalence. Failure of a universal polynomial bound need not give an exponential lower bound, and a universal subexponential bound need not give the polynomial circuit sizes required for TC^1 or NC^2. Familiar targets already in NC, including bounded-depth formulas and polylogarithmic-depth circuit families, cannot establish the stated exponential lower bound because Proposition 11.3 gives subexponential sample bounds for them.

Several representation questions remain meaningful between these barriers. Which subclasses of TC^1 have polynomial-product tanh representations? Can a useful quantitative relation be proved between parallel-circuit resources and the smallest product representation? Can one obtain a general learning bound between polynomial and exponential depth dependence without forcing polynomial-size parallel circuits? The present results delimit these questions but do not settle them.

A Constant-width bounded-weight teachers can encode every small circuit

The teacher width in Corollary 10.2 need not grow. The following finite-horizon construction makes this precise. The idea of encoding discrete memory in real registers is classical: saturated-linear neural simulations use base encodings [5], and Kilian and Siegelmann [6] establish sigmoidal universality. We give an independent finite feedforward construction with explicit depth and error bounds. In particular, we do not infer a polynomial-time tanh simulation from an earlier universality theorem.

Theorem A.1 (A constant-width tanh compiler)**.** *Let $d \geq 1$, and suppose an acyclic Boolean circuit with S gates, each of fan-in at most two, computes $f : X_d \to \{-1, 1\}$. There is a tanh network with augmented input $(1, x)$, no separate hidden biases, hidden width 24, at most*

$$H \leq 5000(d + S + 1)^3$$

hidden layers, every physical coefficient of absolute value at most one, and scalar linear readout satisfying $f(x)z(x) > 1$ for every $x \in X_d$. All coefficients may be chosen dyadic with $O(H + \log(d+1))$ bits each. The construction only guarantees a Frobenius product bounded by $\exp(O((d + S + 1)^3))$; it is not a polynomial-product representation.

A.1 Registers and stable amplification

Put $n = d + S$, $M = n + 1$, and set

$$c = \tanh 1, \quad \kappa = c^{-1}, \quad \delta = \eta = 4^{-20M}, \quad \alpha = \frac{\operatorname{arctanh} \delta}{\delta}, \quad K = 500M.$$

We first use six logical registers R, T, U, B, C, D. The last two are constant rails, initialized from the augmented input by preactivations 1 and $\operatorname{arctanh} \delta$. The updates

$$C \leftarrow \tanh(\kappa C), \qquad D \leftarrow \tanh(\alpha D)$$

keep them exactly equal to c and δ. Here $0.761 < c < 0.762$ and $\kappa, \alpha < 2$. The D rail is included to make tiny constants explicit; the particular memory updates below can also use tiny coefficients multiplying C.

Let $\varphi(u) = \tanh(\kappa u)$. Its fixed points include $0, \pm c$. Concavity gives

$$\frac{\varphi(u)}{u} \geq \frac{\tanh(3/4)}{3c/4} > 1.1 \quad (0 < u \leq 3c/4),$$

and on the invariant interval $[3c/4, 1]$,

$$|\varphi'(u)| \leq \frac{\operatorname{sech}^2(3/4)}{c} < 0.8.$$

By oddness the same bounds hold with signs reversed. Thus, whenever $\delta/10 \leq |u| \leq 1$, K iterations give

$$|\varphi^K(u) - c \operatorname{sign} u| \leq \eta.$$

Indeed it suffices to take

$$\left\lceil \frac{\log(8/\delta)}{\log 1.1} \right\rceil + \left\lceil \frac{\log(1/\eta)}{\log 1.25} \right\rceil < 500M$$

iterations. Once a register is within η of $\pm c$, applying φ keeps it there. We call this a robust bit register. Small memory registers instead use the hold update $u \leftarrow \tanh u$. Every layer updates all registers; the schedule below specifies which registers use a small hold, a robust hold, a reset, or an arithmetic update.

A.2 Base-four memory and a fixed read schedule

Store bits $b_1, \ldots, b_m \in \{-1, 1\}$, with $m \leq n$, as

$$r^* = \delta \sum_{i=1}^{m} b_i 4^{-i}.$$

Then $|r^*| \leq \delta/3$ and, for a nonempty word, $|r^*| \geq \delta/6$, with sign equal to b_1. The ideal operations are

$$\text{pop}: \quad r^* \mapsto 4r^* - \delta b_1, \qquad \text{insert}: \quad r^* \mapsto (r^* + \delta b)/4.$$

Initialize

$$R = \tanh\left(\delta \sum_{i=1}^{d} x_i 4^{-i}\right), \qquad T = U = B = 0.$$

The initial memory error is at most δ^3.

To read position p of R, first set $T \leftarrow \tanh R$. For $i = 1, \dots, p$, set $U \leftarrow \tanh T$ and apply K amplifier steps to U. If $i < p$, pop using

$$T \leftarrow \tanh\bigl(4T - (\delta/c)U\bigr).$$

Throughout the read, hold R in its small mode. The error bound below shows that each amplification reads the intended bit, so at completion U is within η of c times the requested bit. Circuit addresses and gate order are known in advance: there is no input-dependent control flow. A read uses at most $1 + n(K + 2)$ layers.

For two operand bits, read the first and save it by $B \leftarrow \varphi(U)$. Hold B robustly while reading the second into U. A NAND operation has update

$$U \leftarrow \tanh(C - B - U),$$

followed by K amplifier layers. Its ideal preactivation values are $-c, c, 3c$; the input error is at most 2η. Hence its sign is the NAND bit with a fixed margin, and amplification restores η accuracy. Insert its output by

$$R \leftarrow \tanh\bigl((R + (\delta/c)U)/4\bigr).$$

Only outputs of the original circuit gates are inserted, so the word never exceeds $d + S = n$ digits.

For completeness, an arbitrary two-input gate needs at most five NAND operations on B, U, T. During this finite program T is a robust bit register; the next read overwrites it. In the following table, $\mathsf{N}(X, Y)$ denotes NAND followed by K amplification steps; unwritten occupied robust registers are held by φ. The initial operands are $B = a, U = b$, and the output is in U. Truth strings are ordered $00, 01, 10, 11$, where 0 represents -1.

Truth string	Sequential assignments
0000	$T = \mathsf{N}(B,B);\ U = \mathsf{N}(B,T);\ U = \mathsf{N}(U,U)$
0001	$U = \mathsf{N}(B,U);\ U = \mathsf{N}(U,U)$
0010	$U = \mathsf{N}(U,U);\ U = \mathsf{N}(B,U);\ U = \mathsf{N}(U,U)$
0011	$U = \mathsf{N}(B,B);\ U = \mathsf{N}(U,U)$
0100	$B = \mathsf{N}(B,B);\ U = \mathsf{N}(B,U);\ U = \mathsf{N}(U,U)$
0101	no assignment
0110	$T = \mathsf{N}(B,U);\ B = \mathsf{N}(B,T);\ U = \mathsf{N}(U,T);\ U = \mathsf{N}(B,U)$
0111	$B = \mathsf{N}(B,B);\ U = \mathsf{N}(U,U);\ U = \mathsf{N}(B,U)$
1000	$B = \mathsf{N}(B,B);\ U = \mathsf{N}(U,U);\ U = \mathsf{N}(B,U);\ U = \mathsf{N}(U,U)$
1001	$T = \mathsf{N}(B,U);\ B = \mathsf{N}(B,T);\ U = \mathsf{N}(U,T);\ U = \mathsf{N}(B,U);\ U = \mathsf{N}(U,U)$
1010	$U = \mathsf{N}(U,U)$
1011	$B = \mathsf{N}(B,B);\ U = \mathsf{N}(B,U)$
1100	$U = \mathsf{N}(B,B)$
1101	$U = \mathsf{N}(U,U);\ U = \mathsf{N}(B,U)$
1110	$U = \mathsf{N}(B,U)$
1111	$T = \mathsf{N}(B,B);\ U = \mathsf{N}(B,T)$

A repeated source uses coefficient -2, within the bound four on logical matrix entries. Literal constants can replace a read by $U \leftarrow \varphi(\pm C)$; no extra memory digit is needed. Two reads, a save, five NAND operations, and insertion cost at most $(2n + 5)K + 4n + 9$ layers per original gate. Including initialization and an optional final read of an original input, the total is at most

$$(2n^2 + 6n)K + 4n^2 + 11n + 2 < 5000(n + 1)^3 =: T_{\max}.$$

Gates outside the designated output's ancestry may first be deleted, and the remaining gates topologically ordered with the output last. Constant circuits and an output equal to an original input are covered by the direct constant assignment and optional read respectively.

A.3 Uniform error bounds

For every real z,

$$|\tanh z - z| \le |z|^3/3,$$

because $z - \tanh z$ is the integral of $\tanh^2$ and $|\tanh z| \le |z|$. Whenever a small register is within $\delta/100$ of its ideal value, its magnitude is below $\delta/2$. A hold or copy then introduces at most δ^3 error. A valid insertion satisfies

$$e_R^{\text{new}} \le e_R/4 + \delta\eta/2 + \delta^3.$$

Dropping the beneficial factor $1/4$, the main-memory error is at most

$$E_R = (T_{\max} + 1)\delta^3 + n\delta\eta.$$

For a temporary read, a pop satisfies

$$e_T^{\text{new}} \le 4e_T + 2\delta\eta + \delta^3.$$

There are at most n pops, and $T_{\max}$ bounds all intervening holds. Consequently

$$E_T = 4^n\big[(2T_{\max} + 2)\delta^3 + 3n\delta\eta\big]$$

bounds every temporary-memory error. With $M = n + 1$,

$$\frac{E_T}{\delta} \le (10000M^3 + 2)4^{-39M} + 3M4^{-19M} < \frac{1}{100}.$$

For example, $M \le 2^M$, $M^3 \le 8^M$, and $M \ge 2$ give the last inequality with ample slack. Also $E_R \le E_T$.

These estimates close a simultaneous induction over the fixed schedule. At extraction, the ideal memory has magnitude at least $\delta/6$, its error is below $\delta/100$, and the initialization $U \leftarrow \tanh T$ introduces at most δ^3 error. Thus its sign is correct and its magnitude exceeds $\delta/10$. The amplifier estimate applies, which in turn justifies the pop and insertion recurrences. The robust NAND computation has the separate constant-scale sign margin established above. Resetting T at each read prevents its intervening robust mode from carrying small-memory error into that read. Every estimate is uniform on the entire Boolean cube, without a probability or genericity assumption.

A.4 Physical width, coefficient size, and finite precision

Every logical layer is a linear map on six registers followed by tanh, with matrix entries bounded by four. Replace each logical register by four identical neurons. Replace an entry A_{ij} by $A_{ij}/4$ on every source-clone/destination-clone edge. Each destination clone then has exactly the logical preactivation, and every physical coefficient is bounded by one. The four initial clones use the initial row directly; all its coefficients are already at most one. This gives width 24 without any separate hidden bias.

The final U is within η of $cf(x)$. Read out $2U/c$, using coefficient $1/(2c) < 1$ on each of its four clones. The signed output is at least $2(1 - \eta/c) > 1.9$. Uniform cloning preserves each internal Frobenius norm: the cloned matrix is $A \otimes (J_4/4)$ and $\|J_4/4\|_F = 1$. The first norm is at most three, the readout norm below two, and each internal norm at most 24. Thus the product is at most $6\,24^{T_{\max}}$, with no uncharged norm reduction.

Exact transcendental rails are unnecessary for the conclusion. Round every physical coefficient toward zero to dyadic error at most ξ, preserving its bound one. With H hidden layers, maximum coordinate errors obey

$$e_1 \le (d+1)\xi, \qquad e_j \le 24e_{j-1} + 24\xi.$$

The resulting readout error is at most $(d+3)24^{H+1}\xi$. Taking

$$\xi \le \left[20(d+3)24^{H+1}\right]^{-1}$$

changes every output by at most $1/20$. The signed margin remains greater than one, and $O(H + \log(d+1))$ bits per coefficient suffice. This proves Theorem A.1.

Corollary A.2 (Fixed-width teachers and the learning barrier)**.** *Boolean families represented with global unit margin by width-24 tanh networks of polynomial depth and coefficient magnitudes at most one are exactly* P/poly*. Consequently, the joint-polynomial-learning hypothesis of Corollary 10.2 already implies* $\mathsf{P}/\mathrm{poly} = \mathsf{TC}^1$ *when restricted to these teachers, provided the learner guarantee still holds at every width* $N \ge 24$*.*

Proof. The compiler gives the inclusion of P/poly. The reverse inclusion is Lemma 11.2, with constant width and weight bound. The compiler has polynomial depth and can be padded to at least d hidden layers by robust holds before readout. Its fixed width and coefficient bounds remain unchanged. The hypothesized sample count is then polynomial in d, so the proof of Corollary 10.2 applies at a sufficiently large learner width chosen before the input law. □

Remark A.3 (The roles of precision and learner width). The memory uses exponentially small state differences and exponentially fine absolute weight precision, but only polynomially many description bits. It is sensitive to polynomial-scale perturbations and gives no polynomial-product or robust implementation of arbitrary small circuits. At fixed learner width 24, ordinary finite-architecture capacity bounds already give polynomial sample bounds. The collapse conclusion depends on a guarantee uniform over arbitrarily large learner widths, even though the teacher width is fixed.

B Exact proof dependencies and verification scope

The structural product theorem, canonical-feature construction, block-cover argument, finite-event extraction, two-majority reductions, and the preceding constant-width compiler are proved in this manuscript. The statistical applications additionally use target-mass and intrinsic-learning results. The following map specifies those inputs; it is not a claim that a short companion statement replaces its supporting proof.

1. The above-one product ceiling is Theorem `thm:posterior-energy-ceiling` of `unrestricted-primary-posterior-ceiling.tex` in the repository's `works-in-progress/unrestricted-depth/source/` directory. The statement, including its actual finite-scale width threshold, is reproduced as Theorem 5.1 here.

2. The scalar Bellman baseline, small-state bridge, monotonicity, and one-step dissipation are Definition `def:critical-scalar-value`, Lemma `lem:scalar-value-properties`, and equation `eq:scalar-step-dissipation` in `works-in-progress/critical-depth/source/critical-rate-foundations.tex`. Proposition 7.1 gives the precise imported statements.

3. The finite-width Fourier target mass is Corollary `cor:critical-finite-fourier-mass` in `critical-finite-width-channel.tex`, in that same critical-depth source directory. Its seed is constructed in `critical-controlled-fourier-seeds.tex`; the finite channel proof uses bounded-score concentration, conditional Gaussian weight shifts, and binary relative entropy. These are finite-width inputs, not a large-deviation limit substituted for the posterior.

4. The uniform target-mass bound is Proposition `prop:long` and the Fourier-seed section of the companion `uniform-target-classes.tex`. Its scale-change proof is detailed in `nearcritical-target-reuse.md`, Sections 1–6: Gaussian reference change; residual monotonicity; zero-cost score downgrade; the circuit prefix; stable lifting of the scalar path; and finite-width normalized slack. Its critical program inputs are the preceding critical-depth sources. The short-padding proof appears directly in the companion's Section 4.

5. The normalized intrinsic input used for sufficiency is Proposition `prop:intrinsic` of that companion. Its two proofs are `uniform-scale-intrinsic.md` and `uniform-sharp-intrinsic.md`. They combine `finite-width-dual-certificate.md`, the scaled importance compression in `fixed-t-compression.md`, and the scalar gain estimate in `uniform-baseline-gain.md`. The sharper argument uses vanishing readout repair so that it retains the coefficient one on V_*. The finite-witness class is defined by representability on every witness of a prescribed size; it is not assumed that every function outside an arbitrary VC class has a small separating witness.

6. The older bounded-fan-in target upper bounds used in Proposition 11.3 are identified by these source-label pairs:

   - `unrestricted-primary-threshold-circuits.tex`: `cor:threshold-circuit-learning`;
   - `critical-all-width-target-learning.tex`: `cor:critical-all-width-real-threshold-learning`.

   They occur in the respective source directories already specified. The all-prefactor extension instead uses Theorem `thm:threshold` of the uniform companion.

The named companion proof notes should accompany the manuscript whenever its all-prefactor statistical claims are distributed. The repository bibliography below identifies the older public sources; the proof notes provide the additional scale-uniform arguments. The geometric and compression Lean modules associated with this research verify components of the structural argument. They do not constitute a Lean proof of the full posterior converse, its target-mass inputs, the constant-width compiler, or the complexity-class implications. All conditional implications and the open status of unrestricted-depth learning are unchanged by this distinction.

References

[1] ChatGPT-6 Astra, Kaarel Hänni, and Dmitry Vaintrob. *Representation constraints and polynomial-depth learning in Gaussian Bayesian networks*. Research draft, 19 October 2026. Repository: https://github.com/kaarelh/AI-works, directory `neural-network-generalization/works-in-progress/unrestricted-depth/source/`. The imported posterior ceiling is `unrestricted-primary-posterior-ceiling.tex`, theorem `thm:posterior-energy-ceiling`, including its explicit width threshold. The imported target-learning bound is `unrestricted-primary-threshold-circuits.tex`, corollary `cor:threshold-circuit-learning`.

[2] ChatGPT-6 Astra, Kaarel Hänni, and Dmitry Vaintrob. *Depth and learning bounds for critical Bayesian neural networks*. Research draft, 19 October 2026. Repository: https://github.com/kaarelh/AI-works, directory `neural-network-generalization/works-in-progress/critical-depth/source/`. The imported scalar facts are in `critical-rate-foundations.tex`, definition `def:critical-scalar-value`, lemma `lem:scalar-value-properties`, and equation `eq:scalar-step-dissipation`. The imported finite-width Fourier target mass is in `critical-finite-width-channel.tex`, corollary `cor:critical-finite-fourier-mass`. The imported learning bound is `critical-all-width-target-learning.tex`, corollary `cor:critical-all-width-real-threshold-learning`.

[3] *Uniform prior-scale bounds for structured Boolean targets*. Companion research manuscript, 24 October 2026, `uniform-target-classes.tex`. The imported finite-width target mass is Proposition `prop:long`, together with its section on Fourier seeds. The physical scalar-program change of reference and normalized finite-width comparison are detailed in the accompanying research note `nearcritical-target-reuse.md`.

[4] ChatGPT-6 Astra, Kaarel Hänni, and Dmitry Vaintrob. *Mean-field Bayesian circuit learning at every width above target complexity*. Research draft, 15 October 2026. Repository: https://github.com/kaarelh/AI-works, directory `neural-network-generalization/main-results/meanfield-all-width/source/`. The cloning cylinder is Proposition `prop:cloning` in `cloning.tex`.

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